

Module 2: Modern Banking

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Lesson 2-1: A Brief History of Banking in the United States

[Lesson 2-0: Overview](#)



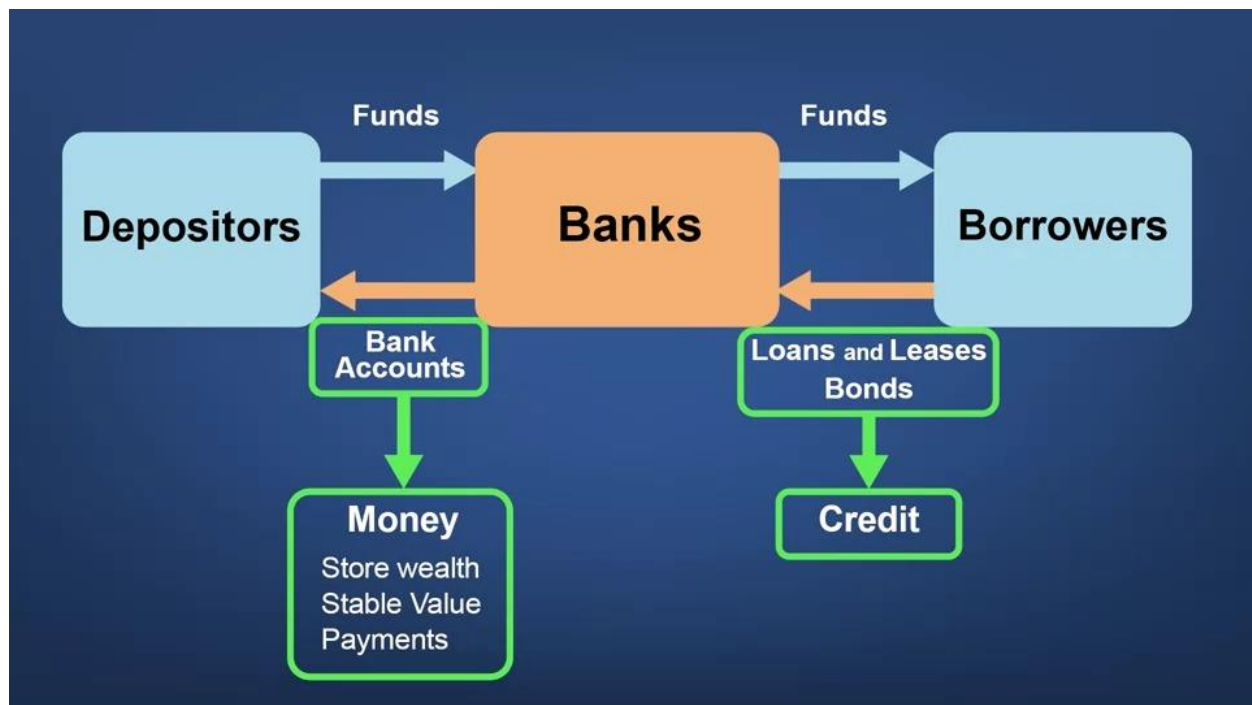
Hello everyone, and welcome to module two of financial institutions. This module is called modern banking. In this module, we're going to zoom in on the business model of modern banks. We will begin with a brief history of money and banking in the United States. If we are going to fully appreciate modern banking, then it is a good idea to understand how we got here. Next, we will describe the organization and functions of modern banks. We will break our modern banks out into their commercial banking and investment banking businesses and examine these business models separately. In a final step, we will go from describing the banking business to analyzing financial statements. We will take a careful look at the balance sheet and income statements of modern banks to understand what's really going on at these financial institutions. By the end of this module, you should understand where modern banks came from, what they do and how they turn a profit.

Lesson 2-1.1: Early Banking in the United States

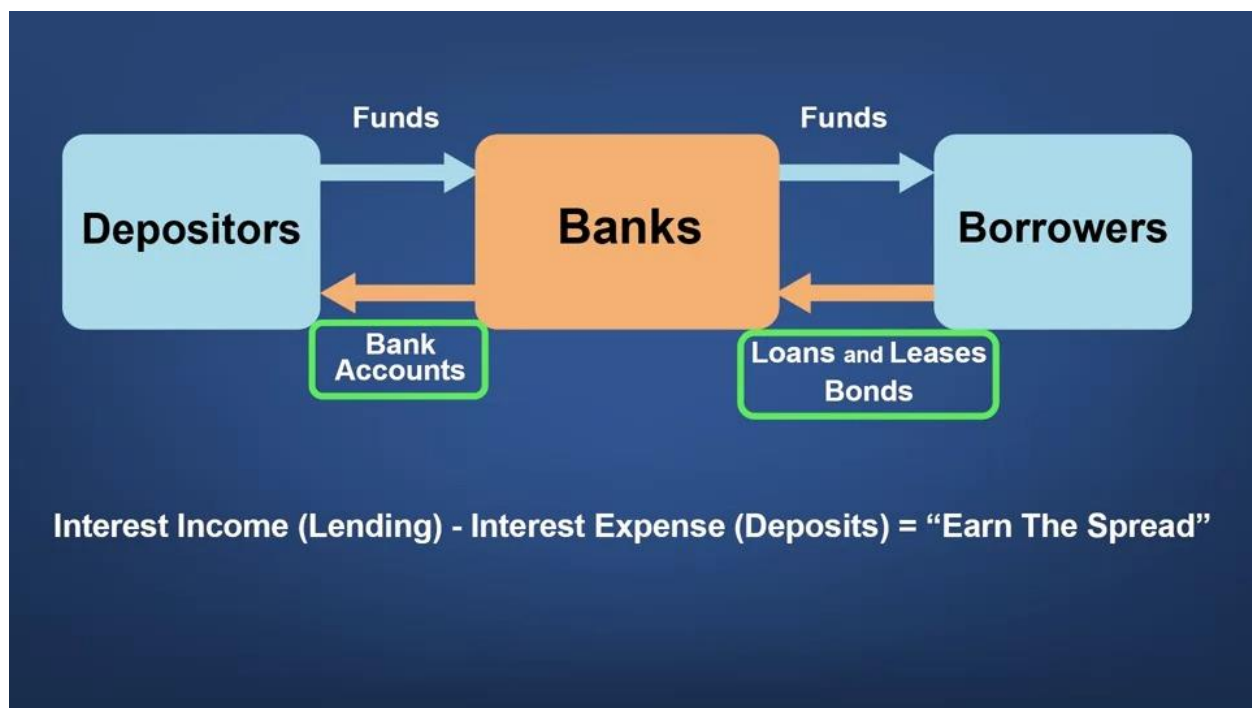
Learning Objectives

- Bank organizational forms
- Historical trends in U.S. banking

What we call the banking system has changed dramatically in the past 100 years. Our parents would probably visit their neighborhood bank and speak to a teller. In case you're wondering, a teller is an actual human being that works at a bank. Nowadays, we barely use ATMs anymore. Most of our banking has either done online or over our smartphones. As the technological and competitive landscape of banking has changed, so too has the structure of the banking industry. By structure, I mean, two things. First, the business model of banks. This includes the types of products they sell. And second, the numbers, sizes and physical locations of banks. In this session, we will focus on the historical trends in the US banking industry. We will see how changes in technology and regulation have given rise to what we call a bank in the 21st century.



Our traditional bank does several things, but we can boil its business model down to taking dollars from depositors and giving them bank accounts and then channeling those funds to borrowers mostly in the form of loans. These bank accounts are basically money. We safely store our wealth without any price risk and we have access to the payment system. For example, we can write checks or use debit cards. Banks also add value by ensuring only credit worthy borrowers get loans.



Traditional banks borrow land and earn the spread between interest income on loans, and interest paid on deposits. They are rewarded for their effort and risk taking.

JPMorgan Chase						
Consumer Businesses			Wholesale Businesses			
Consumer & Community Banking			Corporate & Investment Bank		Commercial Banking	Asset & Wealth Management
Consumer & Business Banking • Consumer Banking • J.P. Morgan Wealth Management • Business Banking	Home Lending • Home Lending, Production • Home Lending Servicing • Real Estate Portfolios	Card & Auto • Credit Card • Auto	Banking • Investment Banking • Wholesale Payments • Lending	Markets & Securities Services • Fixed Income Markets • Equity Markets • Securities Services • Credit Adjustments & Other	• Middle Market Banking • Corporate Client Banking • Commercial Real Estate Banking	• Asset Management • Wealth Management

Modern banking is complicated. Here is JPMorgan Chase's organization charts. It shows the business segments in which JPMorgan operates. This includes the consumer banking business for people like you and me, but also the wholesale banking business which involves a lot more products. Organizations like JPMorgan have tens of thousands of subsidiaries that offer all types of financial services. How did we get from the traditional model of banking to a financial supermarket like JPMorgan Chase?

Organizational Forms

- Unit banking versus branch networks
- Holding companies
- Geographic versus product markets

In this session, we will discuss three organizational forms. The unit bank, the branch network and the bank holding company. Two themes will run through our discussion. First, where banks are legally permitted to operate. And second, the types of financial products and services that these banks are allowed to sell. In the 21st century, it seems like banks are able to operate wherever they want and offer every imaginable financial service. However, it is important to recognize that this was not always the case. Throughout history, giving banks permission to expand or change what they do has been a source of vigorous policy debate.

Early Banking in the U.S.

- Unit banking versus branch networks

Let's start with how banks spread themselves out geographically. By this, I mean, the number and locations of the brick and mortar office spaces of a bank. In the past, there were many restrictions on the number of branches that banks were allowed to have. States differed in their approaches. Some states were very strict. Illinois only permitted unit banking. This meant that a bank could only have one physical location.



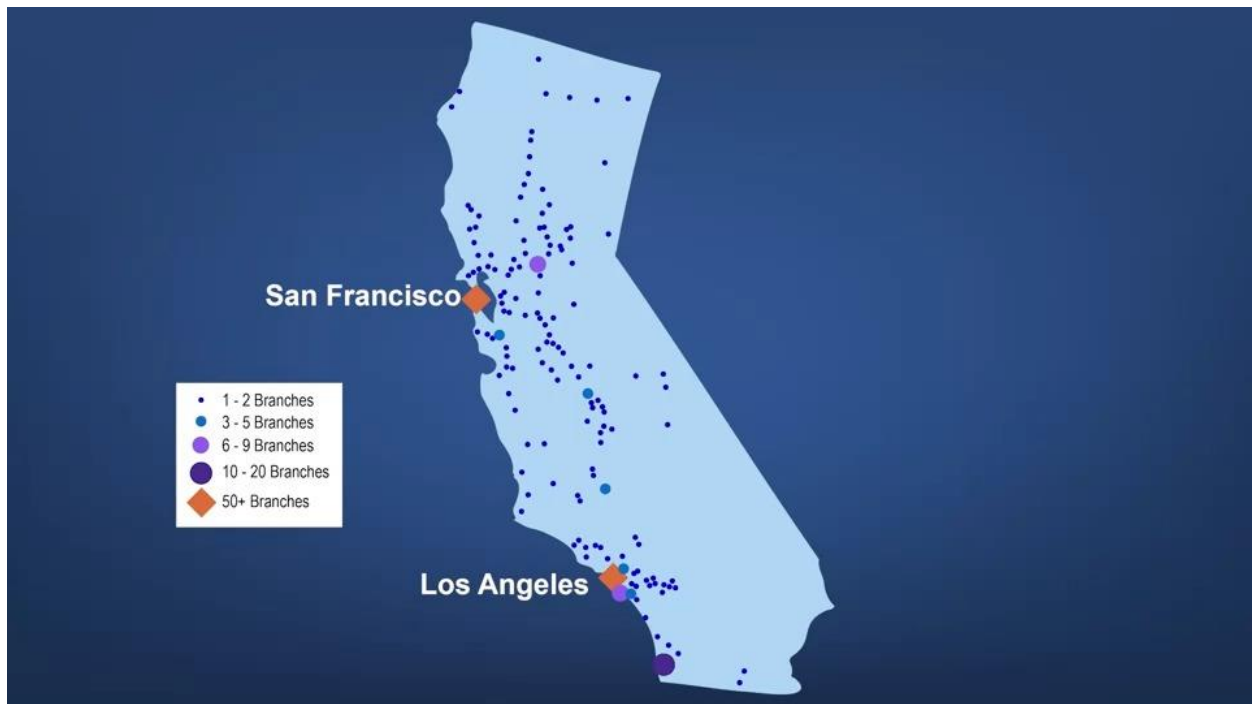
Here is the Continental Illinois bank. It was located At 28 South LaSalle Street in the Financial District of Chicago. All of the employees including the CEO were in one place.

All of the loans were recorded at this building. Customers knew where to go. Unit banking was the law of the land in Illinois from 1870 until as late as 1985. I was shocked the first time I heard that.

Early Banking in the U.S.

- Unit banking versus branch networks
- Illinois versus California

California was at the opposite end of the spectrum. They have always allowed banks to have branch networks within the boundaries of the state.



Here is the Branch Network of Bank of America, circa 1929 used to be headquartered in San Francisco with branches throughout the state.

Early Banking in the U.S.

- Unit banking versus branch networks
- IL versus CA
- Why?

There are several reasons why Illinois and other states enacted strict branching rules. Opponents of bank branching argued that banks could grow too large and exert monopoly power. Bankers, especially those in small towns were strong supporters of branch restrictions. They believe it protected them from competition from larger out of town banks. Other states believe that larger, more geographically diversified banks would be more resilient and safer. While there is much disagreement about whether unit banking or branch networks were safer, banking panics were common in this era. Here are two famous quotes about this time.

History of Banking Panics

“Since 1793 panics have occurred [in the United States] in the following years: 1797, 1811, 1813, 1816, 1819, 1825, 1837, 1847, 1857, 1866, 1873, 1884, 1890, and 1893. The true name for a system with this record is the panic system.”

Theodore Gilman, Federal Clearing Houses, 1899
Reprinted in Gary B. Gorton, Misunderstanding Financial Crises, 2012

History of Banking Panics

“The Panic of 1907 was the first worldwide financial crisis of the twentieth century.”

Moen and Tallman, Federal Reserve History, 2015

The first points out how frequent banking panics were back then and the second points out how they eventually became global in scope. Panics work like this. Typically, some depositors began to worry about losing their savings. Why did this happen? Well, these depositors began to worry that the economy might be taking a turn for the worse and their bank might incur large losses on their loans and eventually fail. Remember that this was before we had deposit insurance. This means that only those depositors who failed to withdraw their money in time would lose out. Hence the run on the bank. These banking panics peaked during the Great Depression of the 1930s. This was probably

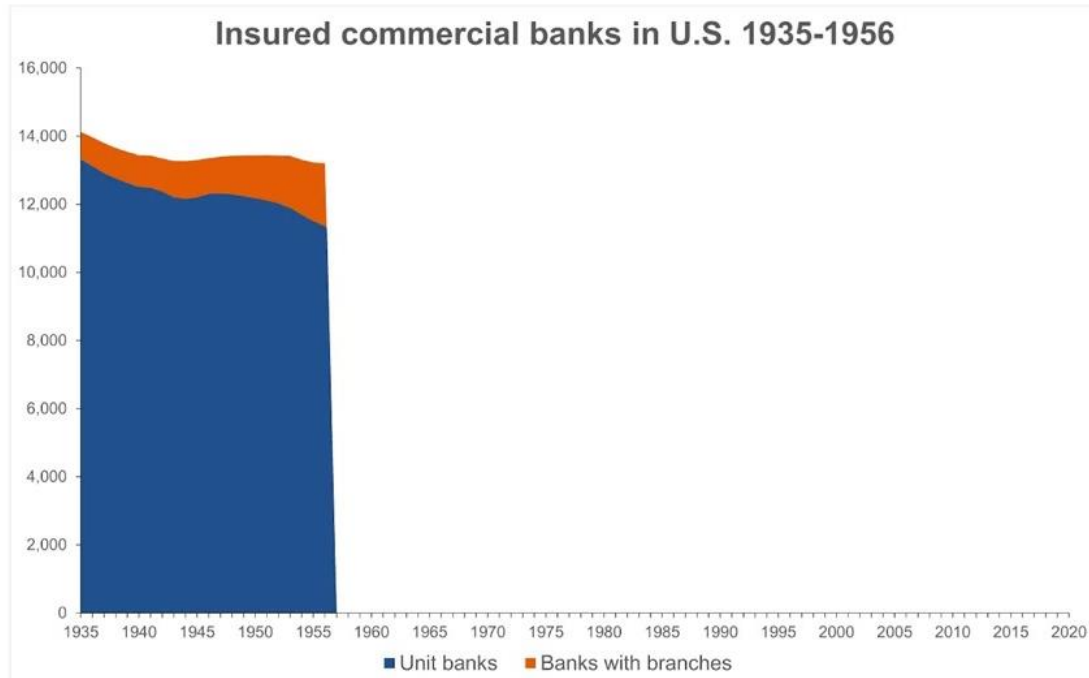
the worst global financial crisis in modern history. More than one-third of all banks failed in the United States, millions of families lost their life savings and the economy crashed.

Regulatory Response to Great Depression

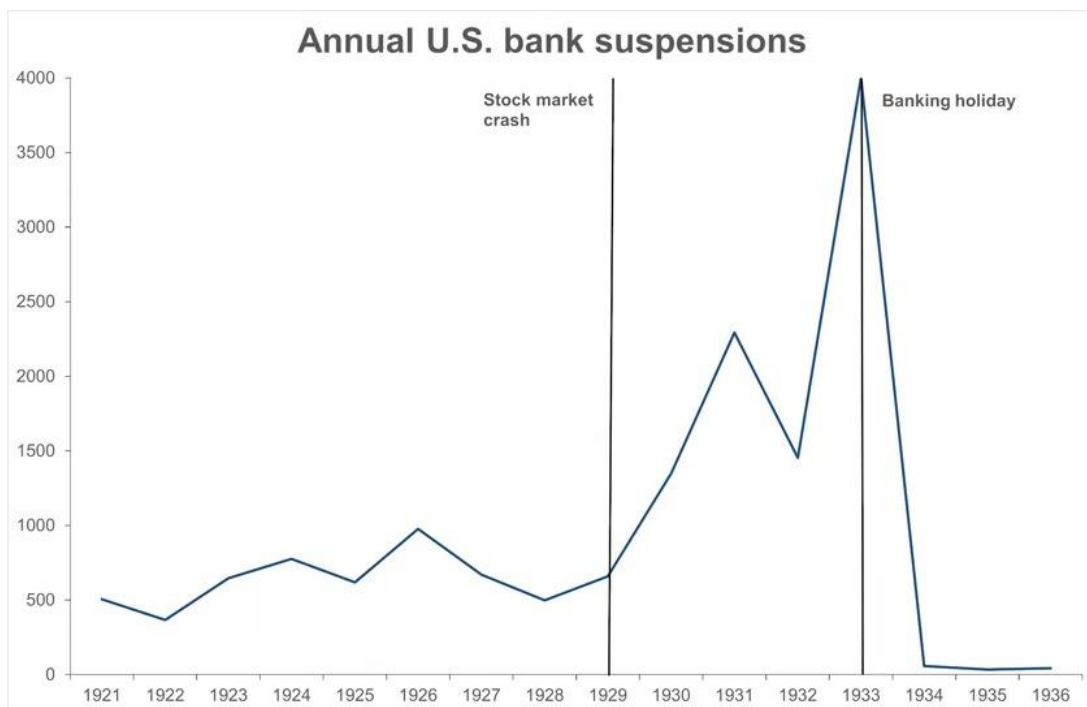
Glass-Steagall Act

- Banned interstate branching
- Banned investment banking
- Banned interest on deposits (Reg Q)
- Established FDIC

As an emergency response to the banking crisis and ensuing depression, President Franklin Roosevelt enacted the Banking Act of 1933. This famous piece of legislation is also known as the Glass-Steagall Act. What changes were put in place. There was a ban on interstate bank branching. This was a geographical restriction on banking. Banks were not permitted to have subsidiaries across state borders. There was a ban on investment banking. Commercial banks were banned from having underwriting or from dealing in private securities, or even writing insurance. The law included regulation Q, which banned interest on deposit accounts. Finally, the act created the Federal Deposit Insurance Corporation. The FDIC provided insurance to individual depositors if their bank failed, they would not lose their savings. These bands reflected beliefs about the causes of bank panics. Interstate branching caused panics to spread like wildfire. Investment banking was too risky. The payment of interest on deposits led to excessive competition among banks. Bank runs were caused by confused investors Worried about losing their savings. The Glass-Steagall Act created a highly regulated and protected commercial banking industry.

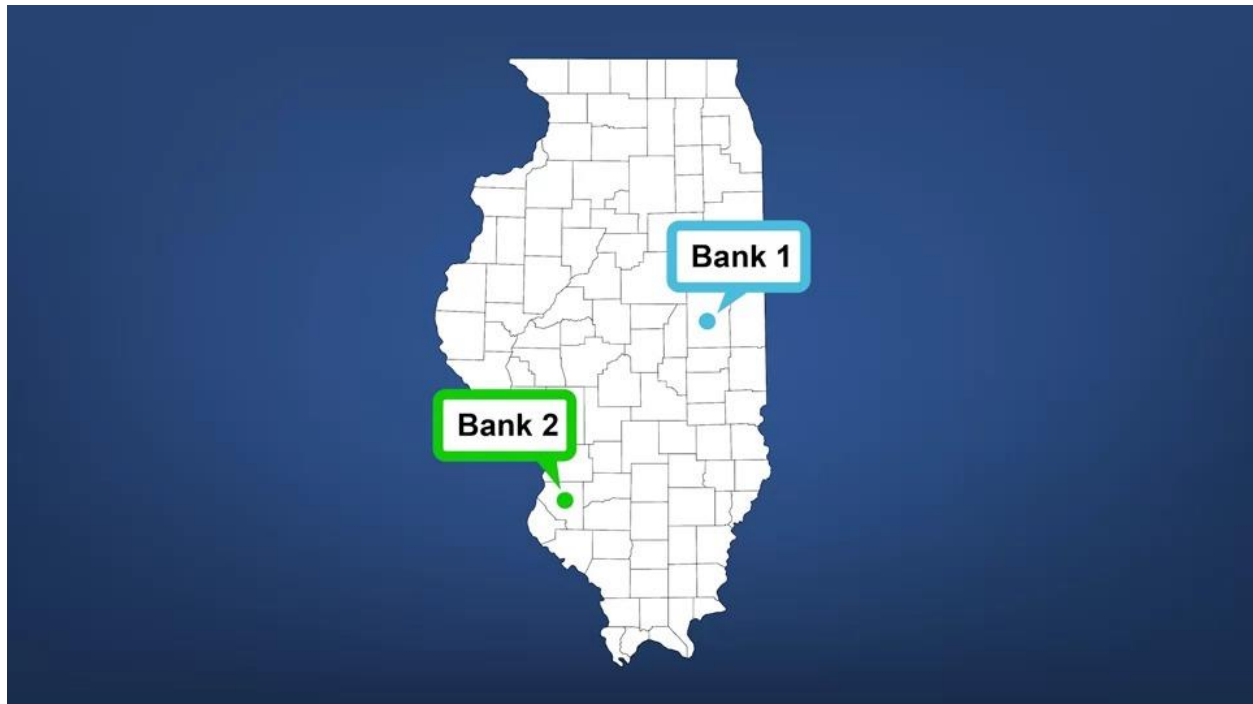


This graph shows the number of insured commercial banks between 1935 and 1956. It shows unit banks and banks with multiple branches. During this period, there were a large and stable number of banks. These were mostly unit banks with local monopolies. This was a system of government insured banks with simple product offerings and limited risk taking opportunities. The Glass-Steagall Act restored confidence in the banking system.

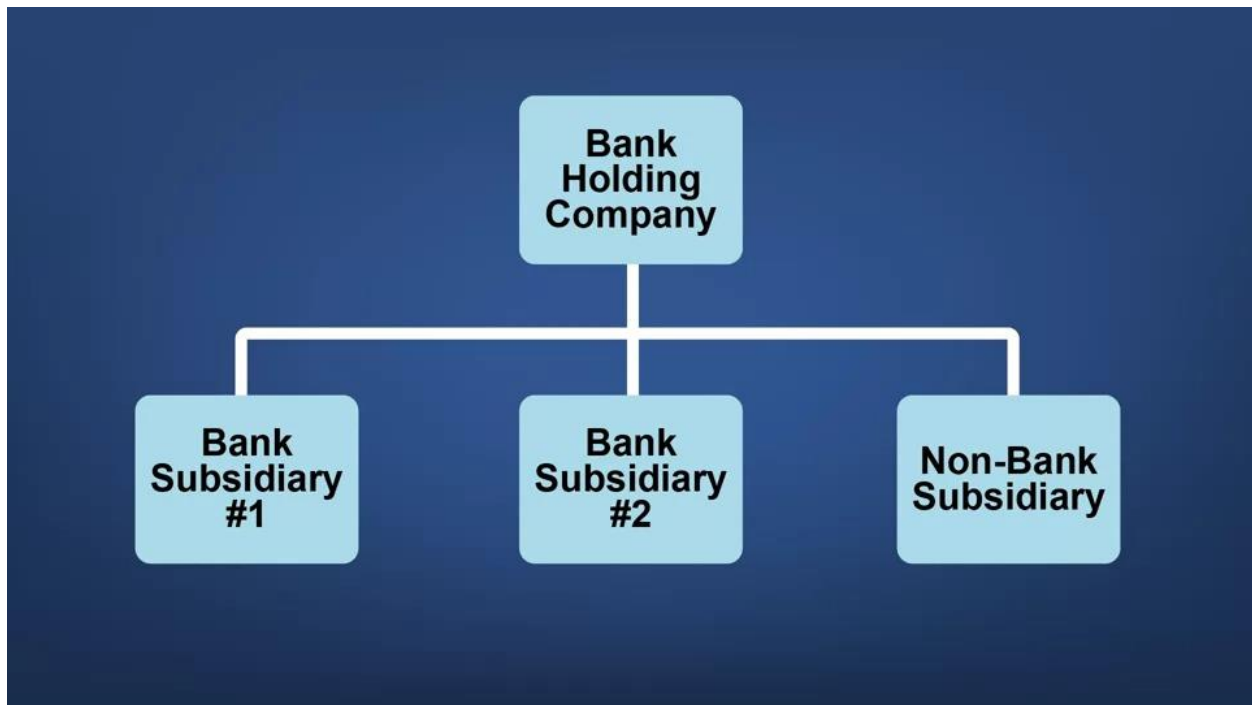


This chart shows bank failures plummeted from a peak of around 4,000 in 1933 to effectively 0. However, at that time, many critics viewed the heavily regulated US banking system as far too fragmented and inefficient.

Lesson 2-1.2: Bank Holding Companies



The next dominant organizational form to emerge was the Bank Holding Company or BHC. A holding company is a shell corporation that owns a group of other firms or subsidiaries. A bank holding company owns one or more bank subsidiaries. For example, let's say it's 1950 and I own a very successful bank in Champaign and I want to expand my operations to Saint Clair County say because I want to geographically diversify. Now Illinois only allows unit banking, so this is impossible. One way to evade the restriction on branching is to first launch an entirely separate bank in Saint Clair and then second I can establish a holding company that acquires both banks. These banks subsidiaries may be considered independent banks and therefore in compliance with the law.



For good measure, why not have my holding company open up a non-bank subsidiary that offers some wealth management services as well. Since this non-bank is not directly affiliated with any bank then I'm probably okay. Right?

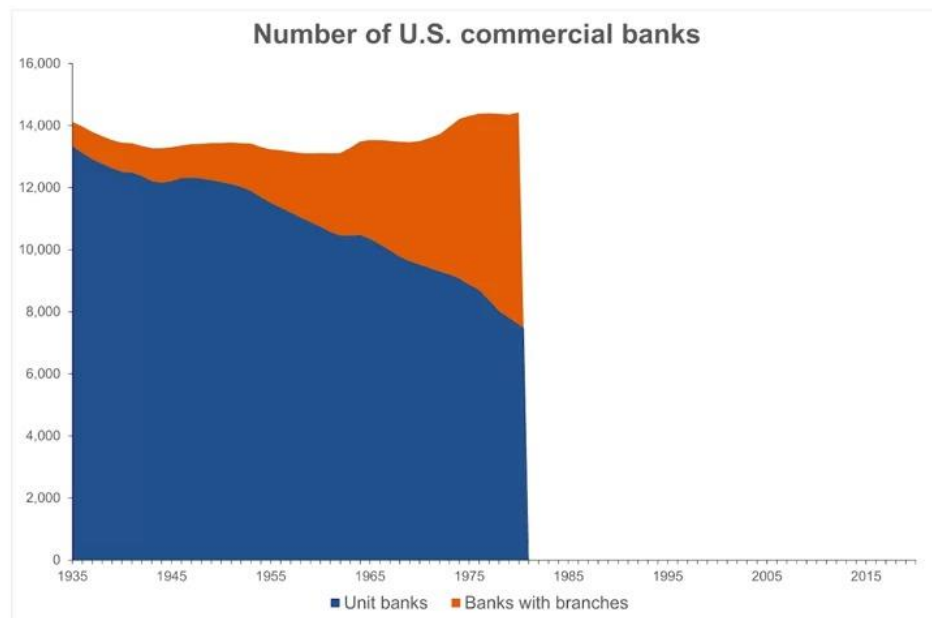
Bank Holding Companies

Regulatory arbitrage

- Evading branch restrictions
- Non-bank subsidiaries

1956 and 1970 BHC Acts

This type of behavior is often called regulatory arbitrage since it exploited loopholes in the law. With the 1956 and 1970 Bank Holding Company Acts, Congress attempted to close these loopholes but by that point the cat was well and truly out of the bag.



I've now expanded on our earlier chart into the 1980s. As it shows by the 1980s, bank branching often via these bank holding companies became the dominant organizational form. Some view the period from 1935 until 1980 as the heyday of US commercial banking. Banks had monopoly power as an industry. This gave rise to the famous 3-6-3 rule of commercial banking.

3-6-3 Rule

“American bankers for decades operated by the 3-6-3 rule: pay depositors 3% interest, lend money at 6% and tee off at the golf course by 3 p.m.

They could afford to be that precise because federal and state laws set the strict rules by which they operated and protected them from competitors.”

Here's a quote, “American bankers for decades operated by the 3-6-3 rule, pay depositors three percent interest, lend money at six percent, and tee off at the golf

course by 3.00 PM. They could afford to be that precise because federal and state laws set the strict rules by which they operated and protected them from competitors”.

3-6-3 Rule

“As a result, the power and prestige of bankers remained as secure as their vaults, while profits were steady and certain.”

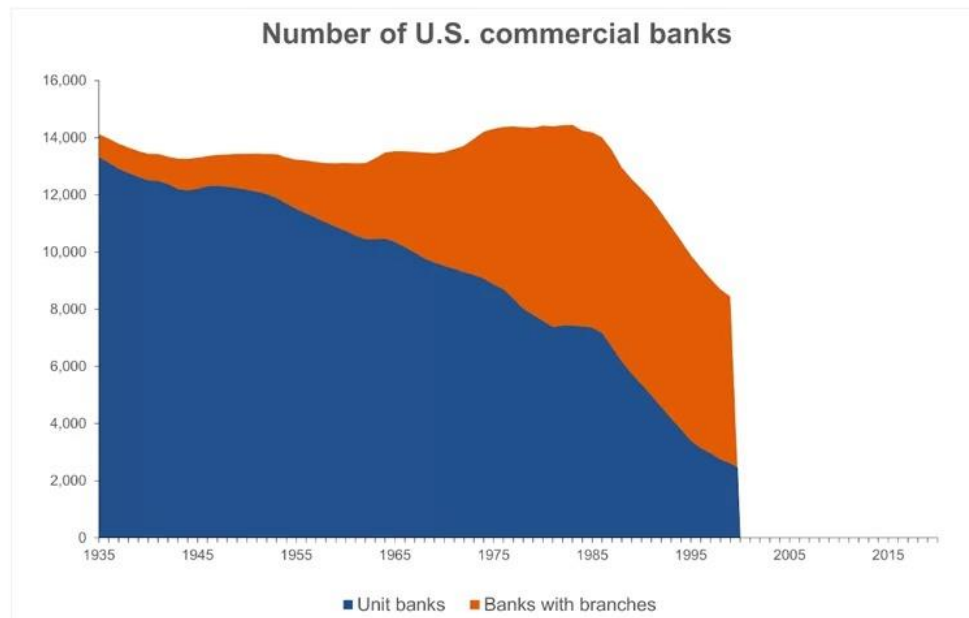
“As a result the power and prestige of bankers remained as secure as their vaults while the profits were steady and certain.” However, all of that began to change beginning in the 1970s and '80s. Suddenly all that's gone.

3-6-3 Rule

“Suddenly, all that is gone. Bankers now face their most strenuous survival test since the Great Depression.”

Time Magazine, December 3, 1984

Bankers now face the most strenuous survival test since the Great Depression.



Back to our chart. The number of unique commercial banks in the United States peaked around 1985. After that there was a collapse in the number of separate institutions. This was a period of dramatic consolidation by 1999, the number of banks had roughly halved to around 8,000.

Causes of Consolidation

- Communication technology
- Substitutes for bank deposits
- Substitutes for bank loans

So what were the main causes of this consolidation? First, technology. Beginning in the 1970s the local nature of banking became less important as mail, telephones, fax

machines, and eventually the Internet became prevalent. We lost the need to walk into a neighborhood bank and speak to a loan officer in person. Nowadays, we can manage our deposits, debit card, mortgage payment, all of that stuff from our smart phones. I can get a loan online from a bank on the other side of the country. All this began in the 1970s. The second major shift came from financial innovations in debt markets. This was an assault on traditional banks in terms of their ability to raise deposits and make loans. Substitutes for deposits arrived. As we mentioned before, under-regulation queue banks were limited in their ability to pay interest on deposit accounts. Beginning in the 1980s, money market mutual funds came on the scene. These mutual funds are non-banks that also offer deposit like accounts, super safe, very convenient, but these funds are not subject to regulation queue. Substitutes for bank loans also arrived. In the 1980s there was seismic shifts in public bond markets. In terms of corporate finance short-term commercial paper and long-term junk bond markets emerged and then securitization which created a brand new funding model for bank loans. This caused further disruption for the banking industry. These were the major sources of competitive pressure for banks. Local banking monopolies were disrupted. This force banks to make hard choices, go out of business, scale up organically or scale up via merger or acquisition. By expanding banks could try and exploit economies of scale and economies of scope this usually involves eliminating redundant costs.

Lesson 2-1.3: Deregulation and Modern Banking

Two Waves of Deregulation

Geographic market

- 1994 Riegle-Neal Act

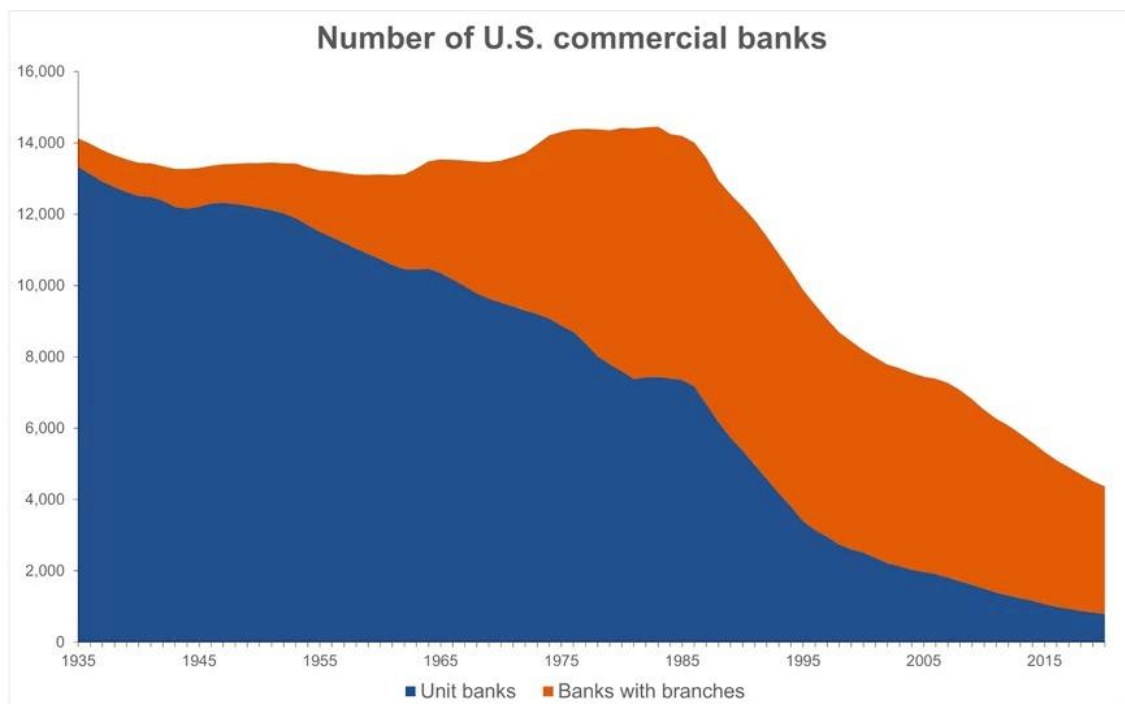
Product market

- 1999 Gramm-Leach-Bliley Act

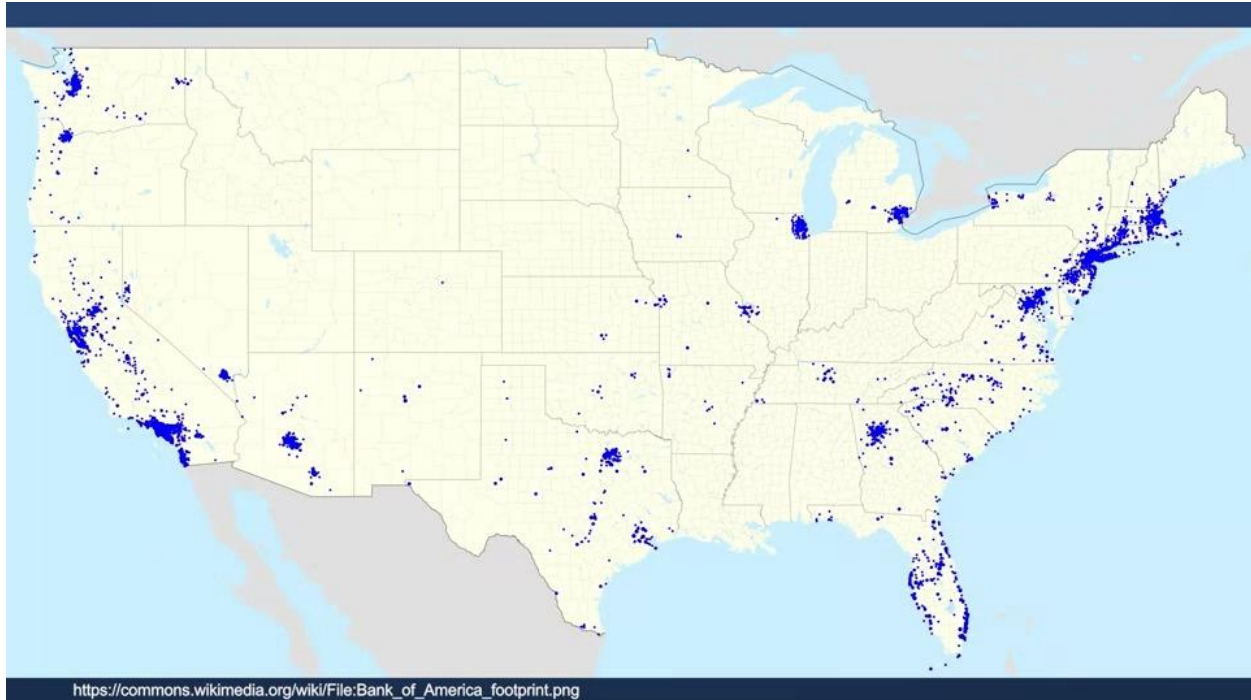
This period of competition led to a paradigm shift in bank regulation. Lawmakers and regulators became less worried about excessive bank risk taking or monopolistic banks. They became more worried about banks closing down because they were unable to compete. Two major pieces of legislation modernized the commercial banking industry. Both were signed into law by President Bill Clinton. These deregulations were designed to help banks increase their efficiency by exploiting economies of scale and economies of scope. First came the Riegle-Neal Act. This allowed banks to diversify geographically by operating across state borders. Nowadays, banks are free to expand their operations throughout the United States, say, by acquiring another bank in another state. Unit banking is a thing of the past. Second, we have the Gramm-Leach-Bliley Act. This law repealed the ban on investment banking found in the Glass-Steagall Act of 1933. Now, commercial banking, investment banking, and insurance underwriting could all be done under the same roof. This allows banks to diversify across product lines. Banks are in the business of information. Operating across multiple product lines might lower information costs. For example, if I know that you have a steady income stream arriving in your checking account, then I might be more willing to cut you a deal on your life insurance. Or if you are a business that I have lent to in the past, then I might be well positioned to underwrite your next bond issue. This deregulation kicked off another round of consolidation and reorganization.



This included a wave of mega mergers creating some of the largest banks in the world, JPMorgan Chase, Bank of America, Wells Fargo.



Here is our chart showing the number of US banks one last time. By the end of the time series, we can see branched banks now dominate.



Finally, here we see Bank of America's geographic footprint, this is in 2010. Each dot represents a bank branch. The largest banks now operate just about everywhere in the United States.

Financial Holding Companies

Activities

- Commercial banking
- Investment banking
- Securities firm

Fee-based income

Today, a handful of mega banks provide a broad array of financial services, all under one roof.

These institutions are organized as financial holding companies. They contain thousands of bank and nonbank subsidiaries.

JPMorgan Chase						
Consumer Businesses			Wholesale Businesses			
Consumer & Community Banking			Corporate & Investment Bank		Commercial Banking	Asset & Wealth Management
Consumer & Business Banking	Home Lending	Card & Auto	Banking	Markets & Securities Services	Middle Market Banking	Asset Management
• Consumer Banking • J.P. Morgan Wealth Management • Business Banking	• Home Lending Production • Home Lending Servicing • Real Estate Portfolios	• Credit Card • Auto	• Investment Banking • Wholesale Payments • Lending	• Fixed Income Markets • Equity Markets • Securities Services • Credit Adjustments & Other	• Corporate Client Banking • Commercial Real Estate Banking	• Wealth Management

Here is JPMorgan Chase again. Its main activities include retail and commercial banking, investment banking. It also includes the functions of a securities firm, so broker-dealer activities. There's also asset management. JPMorgan runs a collection of mutual funds and hedge funds through its asset management arm. With this new organizational structure, modern banks have a greater emphasis on fee generation. Investment banks and securities firms earn fees from the services they provide. The 3-6-3 model of banking is a thing of the past.

Summary

- From unit banks to FHCs
- Industry consolidation

Let's wrap up. In this session, we have discussed banking from the early 20th century, all the way through to the present day. Early banks were simple. They focused on deposit taking and lending, often in one physical location. Regulations were strict. Over time, new technology and competition from nonbanks eroded the market position of our traditional banks. A wave of deregulation followed, designed to help our banks modernize. Banks were permitted to expand geographically and into new product markets. This gave rise to the financial holding company structure and the concentrated banking industry that we see today. Was there too much deregulation? Have some mega banks become too big to fail? The largest U.S. banks were central to the 2008 financial crisis, which suggests that the answer might be yes. We will address this complex question and what regulators have done about it at another time.

Lesson 2-2: Organization and Functions of Modern Banks

[Lesson 2-2.1: Bank Types and Borrowing](#)

Commercial Bank

- Definition
- Classification

We interact with banks, probably more often than any other financial institution out there. Banks provide a broad set of financial services. They safely store our money, provide us with access to the payment system. They give us access to financial markets. And they make loans to creditworthy borrowers. The larger banks of financial supermarkets that offer both commercial and investment banking services all under the same roof. In this session, we are going to focus on the business model of commercial banks. We will describe how commercial banks raise funds, how they borrow, and what they do with them, how they lend. Along the way we will make the distinction between retail and wholesale banking. Let's start by defining what a commercial bank is. A commercial bank is a financial institution that takes deposits and makes consumer real estate and business loans. Historically, commercial banks would cater to businesses. That's why they have the commercial label. But nowadays they serve households and individuals as well. We usually classify commercial banks according to their size and function.

Community



Dori, CC BY-SA 3.0 US, via Wikimedia Commons

At one extreme, we have the community banks. These are small banks that serve consumers and small businesses in their communities. They take deposits locally from people like you and me, and they lend these funds out locally also to people like you and me. This is what we call retail banking. In terms of size, they have assets under management. This will mostly be loans under \$1 billion.

Regional



Beyond My Ken, CC BY-SA 4.0, via Wikimedia Commons

Next up, we have the regional or super regional banks. These are larger than community banks. They have assets ranging from tens of billions of dollars to more than

a trillion dollars. They will be headquartered in larger cities and can have offices and branches that run throughout the country. Some are organized as financial holding companies with multiple bank and non-bank subsidiaries. The commercial banking arm of the business operates on a larger scale. We have retail lending, but we also have wholesale lending, which means commercial and industrial lending to large corporate customers. They also have access to wholesale funding markets. This means that in addition to borrowing from retail depositors through branch networks, these banks can also borrow funds in debt markets. I will be more specific about these borrowed funds in just a moment.

Money Center



mattbuck (category), CC BY-SA 2.0, via Wikimedia Commons

This brings us to the money center banks. These are mega banks that rely on wholesale funding to support their lending activities.

Largest US Banks by Total Assets

Rank 2020 Q2	Rank 2020 Q1	Change	Company	City, state	Total assets (Billions)	Total deposits (Billions)
1	1	-	JPMorgan Chase & Co (JPM)	New York, NY	\$3,213.12	\$1,931.03
2	2	-	Bank of America Corp. (BAC)	Charlotte, NC	\$2,741.69	\$1,718.67
3	3	-	Citigroup Inc. (C)	New York, NY	\$2,232.72	\$1,233.66
4	4	-	Wells Fargo & Co. (WFC)	San Francisco, CA	\$1,968.77	\$1,410.71

The US based money center banks are located primarily in New York City, and they sit at the center of the interbank loan and currency exchange markets. Citigroup is a money center bank. Wells Fargo is a mega bank, but it also has the largest branch network in the country. It can easily obtain deposit funding. Wells is a super-regional bank.

Business Model

- “Earn the spread”
- Borrowing?
- Lending?

Commercial banks borrow, lend and earn the spread. They borrow and pay some expenses on that, and then they lend and earn some interest on those investments.

What does this actually mean in practice? How do banks borrow? How do banks lend? Okay, let's start on the funding side. Banks get their funding from savers. They raise deposits from individuals and businesses to convince depositors to lend them money. Banks offer safety and convenience. Banks can also borrow from businesses and other financial institutions through financial markets. This usually requires interest payments.

Funding Sources

Transaction accounts

Demand deposits

Nontransaction accounts

Savings accounts

Small CDs

Other borrowed funds

Large CDs

Interbank borrowing

Repurchase agreements

Commercial paper

There are two types of deposit accounts. Transaction accounts that include demand deposits. These are our checking accounts. They pay no interest, but we can withdraw our funds whenever we want. Non transaction deposits include savings accounts and small certificates of deposit. These are savings instruments that usually offer us some interest payments. In exchange, these deposits are a bit harder to access. We might only be able to make a limited number of withdrawals per month or with a CD. There might be a significant penalty for early redemptions. Borrowing is the second major source of funds, especially for the larger banks. One bank with a funding shortfall can borrow from another bank in the interbank market. This is also called the Federal Funds Market. These are short term unsecured loans. Banks can borrow on a secured basis using a financial instrument called a repurchase agreement or a repo. Repo borrowing works a lot like a pawn shop. Let's say I'd really need a few 100 bucks to buy tickets to the next deadline I basketball game.

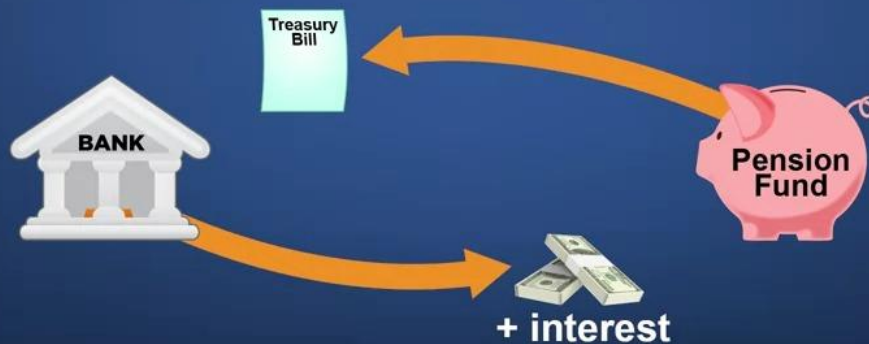


I can take my golf clubs to a pawn shop. We all agree that these clubs are worth \$500. The pawn shop guy will take the clubs and give me 400. This is a \$400 loan secured by the clubs, and then next week after I cash my paycheck, I can buy my golf clubs back for the full \$500, and if I don't show up next week, the pawnshop guy gets to keep my clubs. This is basically how a repurchase agreement works.



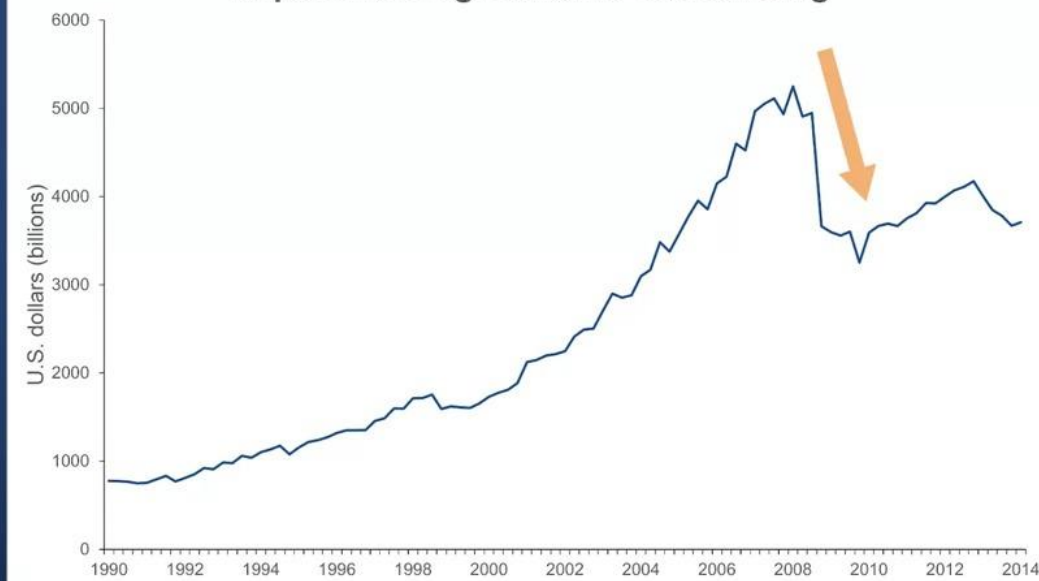
Day Two

Bank repurchases treasury bill from the pension fund for cash plus interest

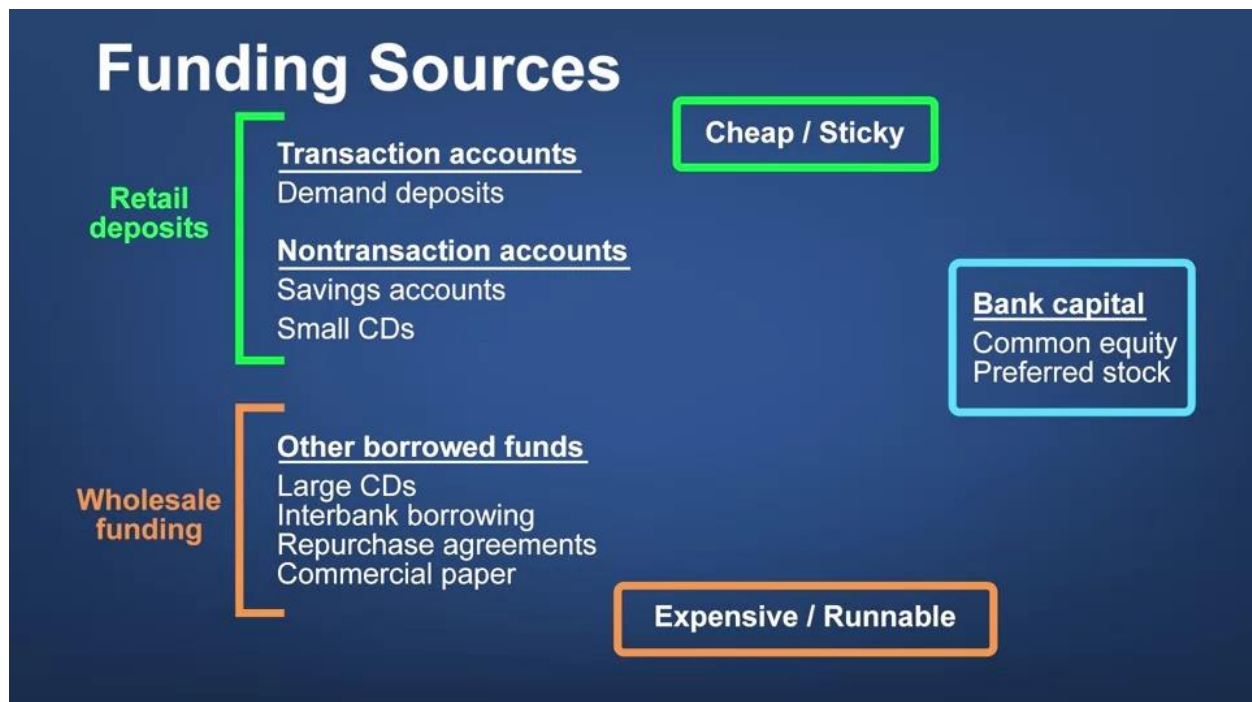


When banks do this, they post high quality securities such as Treasury bills to borrow cash overnight from cash-rich institutional investors such as pension funds.

Repurchase agreements outstanding



The trouble with repos and other short term borrowed funds is that they tend to disappear when investors get worried. It was famously observed that the panic of 2007 was a run on the repo market. The volume of outstanding repos collapsed almost overnight, so some banks lost a really important source of funds. Lehman Brothers was a prime example of this.



We can classify funding sources in terms of how expensive they are and how are unable they are. Retail deposits can require a large infrastructure and a lot of fixed costs to bring in. However, this funding is cheap on the margin, and because of government backed deposit insurance, it tends to stay within the banking system during panics. Wholesale funding is the opposite. Banks have to pay competitive rates to attract it, and during bad times, it tends to run away. Bank capital is the last piece of the funding puzzle. This is the owner's investment in the bank. It mainly consists of common and preferred stock, and just like any business, it includes retained earnings or the profits that are reinvested in the bank. Unlike debt, bank capital never matures, so it is as sticky as it gets. However, because equity is subject to price risk, it is more expensive to raise than deposits.

Lesson 2-2.2: Bank Lending

Lending

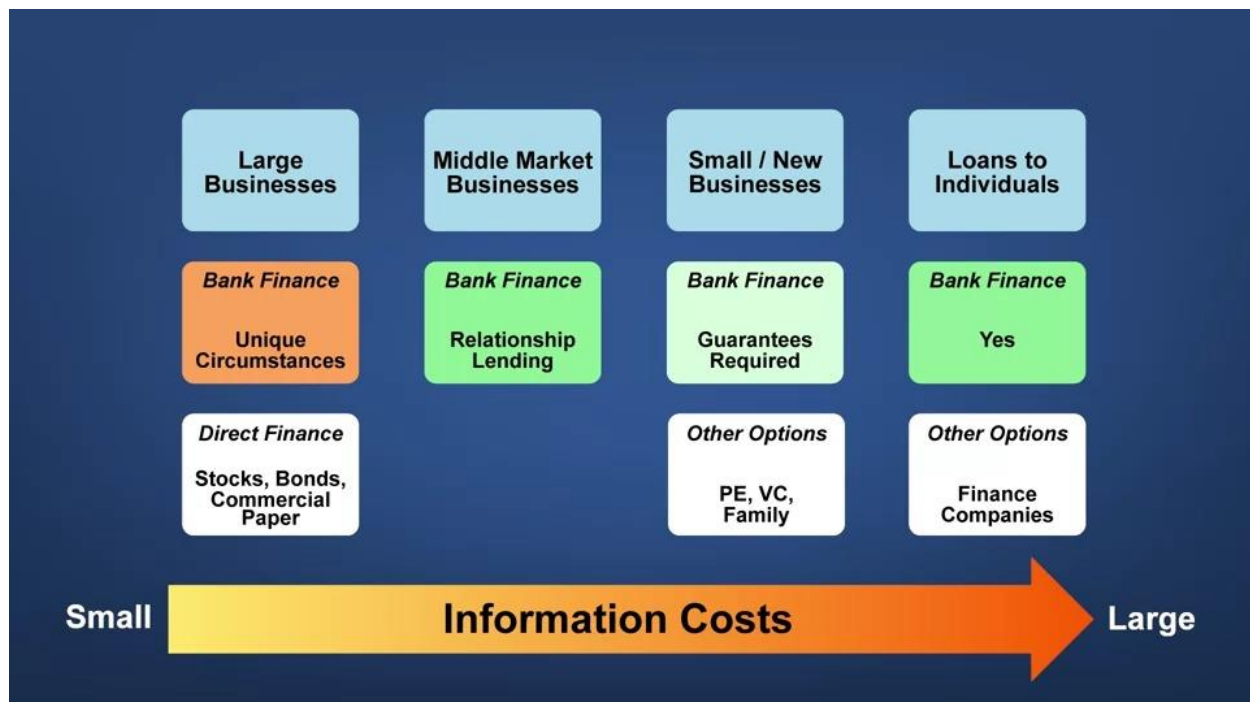
- Loans
- Customers
- Lending process

Let's move on to bank lending. We are going to divide our discussion into three parts. First, we are going to discuss loans. Second, we will run through the types of customers that banks deal with on the lending side. Third, we will talk about the actual mechanics of bank lending. We will describe how a loan comes into existence. Formally, what is a loan? Well, it's a financial instrument, it is debt. When I borrow from a bank I am entering into a legally binding commitment to repay according to some predetermined contractual terms.

Loans

- What is a loan?
 - Rates, collateral, covenants, priority
- Loans to individuals
- Commercial lending

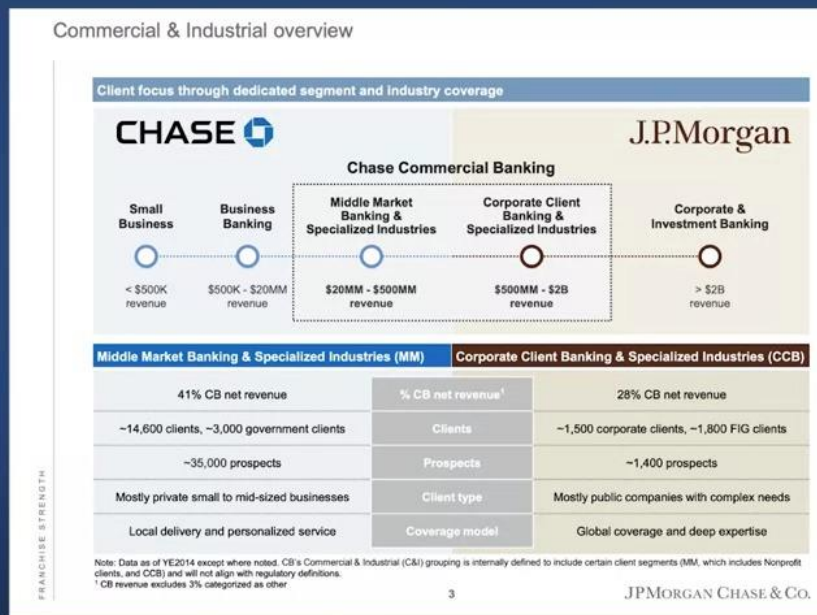
There will be an interest rate. This rate may be variable or it may be fixed. The loan may be secured by some collateral. There might be performance-related clauses written into the contract, these are called covenants. Senior debt gets the first priority on business assets in the case of bankruptcy. Banks usually break their lending business down into two categories: consumer credits and commercial lending. Consumer credit includes loans backed by residential real estate. This includes mortgages and home equity loans. We also have credit cards, auto loans, and student loans. Commercial and industrial lending includes loans to businesses. These can be commercial real estate loans, term loans which include long-term lending for investment, or business lines of credit which act a lot like credit cards for businesses and can be used for working capital.



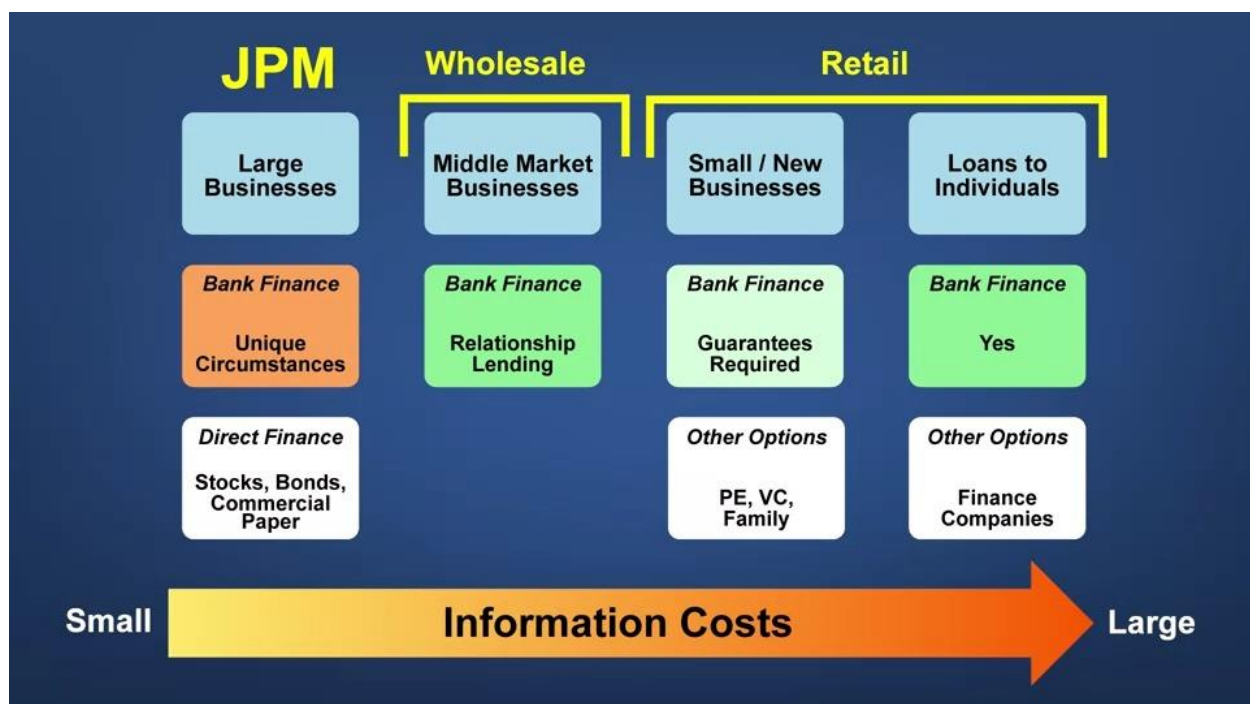
Let's take a look at the customer base of commercial banks. Who do banks lend to? Larger businesses tend to bypass commercial banks and borrow directly from financial markets. They issue bonds and raise equity finance. There is lots of public information about these firms and information costs are small. Sometimes the larger firms will use commercial banks to arrange loans when they are in a special situation. This could be a takeover transaction or a troubled debt restructuring. Middle market lending is the bread and butter of the commercial banking business. This is sometimes called relationship lending. These are medium-sized firms with limited public information. They are probably not listed on a public stock exchange and investors might be worried about adverse selection. Finding your business customers, performing credit analyses, funding new loans, and maintaining relationships. This is the central role of loan officers and credit committees at most banks. Small and new businesses typically do not have access to bank credit without some form of guarantee. These can be personal guarantees of the business owner, the Small Business Administration may partially guarantee these loans. The business owner might pledge his personal assets as collateral. Houses, boats, other financial wealth. Small businesses are risky and often fail. Guarantees and collateral help reduce the risk and bring banks on board. Among loans to individuals, mortgage credit is the core focus of community and regional banks. These loans are also information sensitive and can depend on the individual circumstances of the borrower. Credit cards, student loans, and auto loans, these tend to be small and costly to make on a per loan basis. There are large economies of scale in this market and so larger banks and non-banks that can create big diversified pools of these loans tend to dominate this space.

JPMorgan Chase						
Consumer Businesses			Wholesale Businesses			
Consumer & Community Banking			Corporate & Investment Bank		Commercial Banking	Asset & Wealth Management
Consumer & Business Banking	Home Lending	Card & Auto	Banking	Markets & Securities Services		
• Consumer Banking • J.P. Morgan Wealth Management • Business Banking	• Home Lending Production • Home Lending Servicing • Real Estate Portfolios	• Credit Card • Auto	• Investment Banking • Wholesale Payments • Lending	• Fixed Income Markets • Equity Markets • Securities Services • Credit Adjustments & Other	• Middle Market Banking • Corporate Client Banking • Commercial Real Estate Banking	• Asset Management • Wealth Management

We can see where these different customers show up across JPMorgan Chase's business segments. Their consumer and community banking arm is what I have been calling retail banking. Consumer and business banking means deposit-taking and providing consumer credits and small business loans through the branch network. Home lending covers residential mortgages and home equity loans. The card and auto segment takes care of credit cards and auto loans and leases. This is all for individuals and small businesses. Commercial banking sits within the wholesale banking business. Commercial real estate includes financing for office space, property development, industrial space, and so on. For business banking, JPMorgan separates out its clients into two groups: middle-market and corporate clients.



According to their investor reports, middle-market banking covers small and mid-sized companies with annual revenues between \$200 and \$500 million. Smaller than this means you are in the community banking bucket. Corporate client banking covers large corporations with annual revenues between \$500 and two billion dollars. Bigger than this means that the investment banking arm will take care of you.



So we can map JPMorgan's business model onto our diagram from before. Here is the wholesale business, it covers lending to middle-market firms, and here is the retail

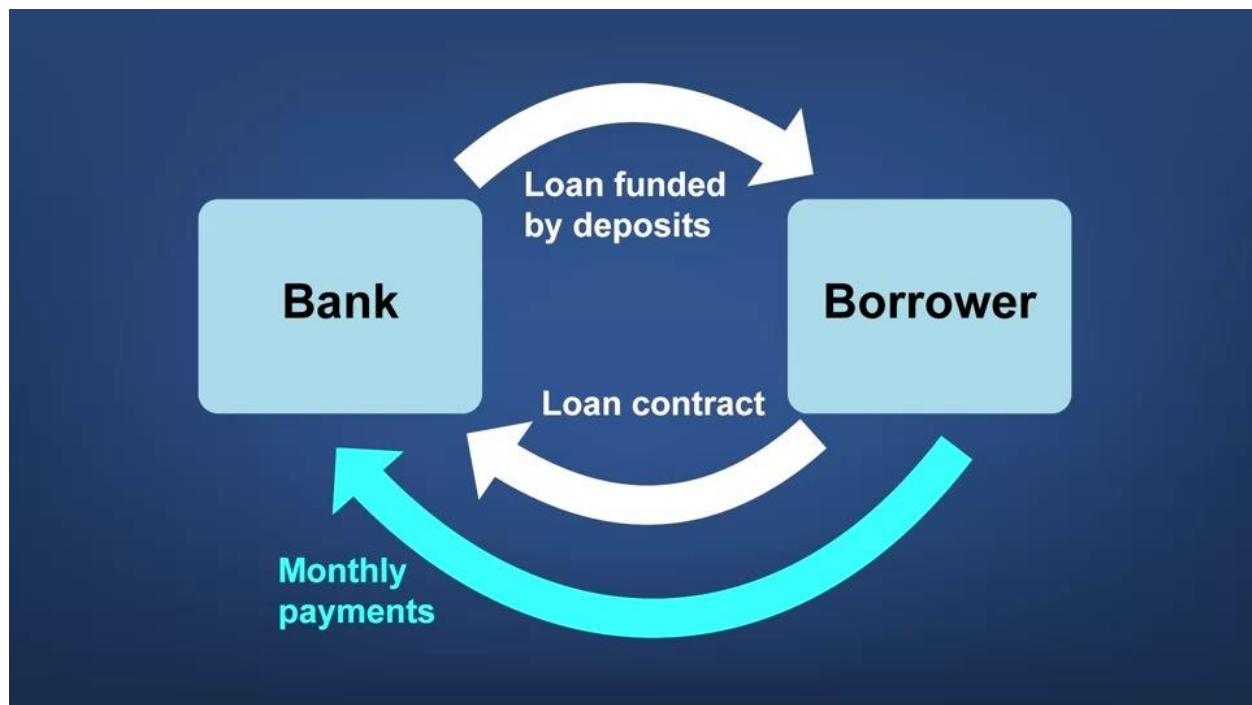
business. It covers lending to small businesses and individuals. OK. That's an overview of the commercial bank business model. Banks raise funds from a variety of sources and grant a broad array of loans.

Lesson 2-2.3: Changes in the Lending Process

Mechanics of Lending

- Origination
- Financing
- Servicing

Before we close our discussion, I want to do two things. First, I want to be a bit more precise about the lending process itself. Second, I want to explain how this lending process has changed in the past 50 years or so. Okay, there are three necessary steps in the lending process. Origination, financing, and servicing. Origination is about granting the loan. A loan officer will meet with a client and determine whether they are credit worthy. This will involve a careful analysis of the borrower's financial situation or business plan. The loan officer will negotiate terms, the loan amount, the maturity rates and fees, collateral and so on. We have called this step screening in our past discussion. Next comes the financing. After committing to lend, the bank will have to come up with the actual funds to give to the borrower. And finally we have servicing, this step involves the bank continuing the relationship with the borrower after the loan has been made. The bank collects payments from the borrower. The bank monitors the borrower's progress and financial condition to maximize the chances that the bank is repaid in full.

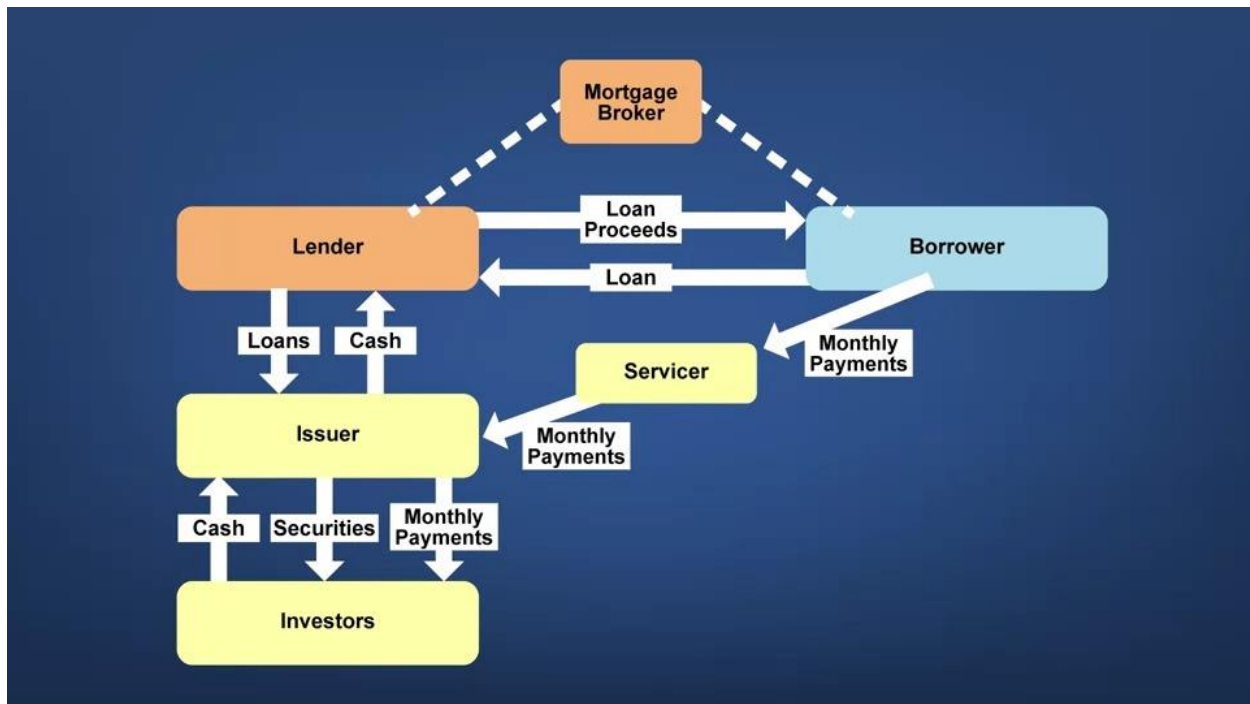


Traditionally, a bank would do all three steps. It grants the loan, it finances the loan users using deposits, and it keeps the loan on the books collecting loan payments along the way until the loan matures. This is what we call the originate and hold model of banking.

Changes in the Process

- Originate and hold
- Originate and distribute

Now, here comes the fun part. In the modern era, each of these three steps can be separated out into what's called the originate and distribute model of banking. Under this business model, commercial banks focus on fee income.



They still meet with the borrower and originate the loan. That part hasn't changed, although sometimes this is done with the help of a broker. In the mortgage market, you might have run into an online broker that connects you with a lender for a fee. Some other financial institution provides the financing that pays for your house. This can be an investor such as an insurance company or there could be an extra step in this process called securitization. The bank may do the servicing, but even this is sometimes outsourced. Rather than making your monthly payment to the bank, you make it to a mortgage servicer who then passes that payment on to the investors. Why do this? Well, if some banks are really good at minimizing information costs and approving the best borrowers for loans, then the theory of comparative advantage tells us that they should focus on that activity. And some other silent partner can come along and provide the dollars.

Changes in the Process

- Originate and hold
- Originate and distribute
- 21st century competition

This is what the originate and distribute model is all about. One final thought. This theory might break down as we arrive in the 21st century. Nowadays, Fintech players like Rocket Mortgage have disrupted the mortgage market. As a consumer, if I am refinancing my mortgage or if I am a great borrower that meets a certain quality threshold, then I can quickly and conveniently get a loan from an online lender. Why do banks have a comparative advantage in this situation? We are seeing similar entry in the business lending space by Amazon and other tech firms. These firms collect huge amounts of information in real time about customer interest and business sales. Perhaps Amazon is better positioned to decide which borrowers should be granted a loan. We shall have to wait and see.

Summary

- Borrowing and lending
- Retail versus wholesale
- Originate and distribute

Okay, let's wrap up. In this session, we took a close look at the commercial banking business. We described how banks borrow and lend. We made the distinction between retail and wholesale banking. Retail banking is about raising funds and lending locally. Wholesale banking is done on a bigger scale. Finally, we introduced the originate and distribute model of banking. This model makes sense if commercial banks have a comparative advantage at selecting the best borrowers.

Lesson 2-2.4: How Are Investment Banks Organized?

Learning Objectives

- Investment bank business model
- Corporate structure

Investment banks provide businesses and other entities with access to financial markets. If you are a business that wants a loan you can go to a commercial bank. If you are a business that wants to sell stocks and bonds you go to an investment bank. If you are a high-net-worth individual that wants to execute a trade, then you might also call your investment banker. This is an oversimplification but it's not too far from the truth. In this session, we want to understand the who, what, when, where, how, why, of investment banking. Our goal is to first define what an investment bank is, and then understand what these financial institutions do and how they're organized. Investment banks connect borrowers and savers directly in financial markets.

Investment Banking

Activities

- Underwriting
- Trading
- M&A advisory

Corporate structure

Investment banks have three main business lines. The first is financing. Investment banks originate, underwrite, and place securities and financial markets on behalf of their customers. These customers are primarily large corporations and governments. Second, since investment banks are already in the business of creating and selling securities, most are involved in securities trading. They execute buy and sell orders for their customers and often trade to make profits for themselves. Investment banks also help to facilitate mergers and acquisitions or other corporate restructuring activities. All of these activities generate fees, underwriting fees, commissions for executing trades, and fees for giving advice. This can be a very profitable business and unlike commercial banking, it does not require long-term funding or extensive branch networks.

JPMorgan Chase						
Consumer Businesses			Wholesale Businesses			
Consumer & Community Banking			Corporate & Investment Bank		Commercial Banking	Asset & Wealth Management
Consumer & Business Banking	Home Lending	Card & Auto	Banking	Markets & Securities Services		
• Consumer Banking • J.P. Morgan Wealth Management • Business Banking	• Home Lending Production • Home Lending Servicing • Real Estate Portfolios	• Credit Card • Auto	• Investment Banking • Wholesale Payments • Lending	• Fixed Income Markets • Equity Markets • Securities Services • Credit Adjustments & Other	• Middle Market Banking • Corporate Client Banking • Commercial Real Estate Banking	• Asset Management • Wealth Management

Let's take a look at the investment banking corporate structure starting with JPMorgan Chase. Ok, here we have JPMorgan's business segments. We can see the corporate and investment bank over here. It sits within the wholesale business arm of the bank. This means that we are dealing with big clients and not retail customers like you or me. The investment bank consists of two businesses – Banking, and Markets. According to their annual report banking means advising on corporate strategy and structure, as well as capital raising and equity, debt markets, and syndicated loan markets. The markets business covers trading in cash and derivative instruments, as well as brokerage services. M&A advisory, securities underwriting, and securities trading. That's the business.

Global Investment Banks FY 2020

% Share	Bank	Revenue (Millions)
9.2	JPMorgan	\$8,410.6
8.2	Goldman Sachs	\$7,536.1
6.7	BofA Securities	\$6,169.3
6.6	Morgan Stanley	\$6,042.1
4.9	Citi	\$4,466.9
3.9	Credit Suisse	\$3,559.4
3.6	Barclays	\$3,292.6
2.4	Deutsche Bank	\$2,201.4
2.3	Jefferies LLC	\$2,071.1
1.9	Wells Fargo Sec.	\$1,714.4
49.5	Subtotal	\$45,463.9
100.0	Total	\$91,847.7

And here is the fees for JPMorgan who is the global market leader. In 2020 they captured about \$8.5 billion of investment banking fees. That is a sizable chunk of their annual revenue. There are many specialized firms in the investment banking industry. Some focus just on brokerage services; other boutique investment banks focus on capital raising for a particular industry. However, we can classify the largest full-service investment banks into three buckets.

JPMORGAN CHASE & CO.

BHC Subsidiary



**Corporate business
Securities trading**

LAZARD

**Standalone
Institutional client base**

First, we have the JPMorgan's of the world. The investment banking arm of JPMorgan is a subsidiary of a commercial bank holding company. These are the largest investment banks and they have a customer base that includes corporations, governments, financial institutions, and high-net-worth individuals. Second, we have investment banks that tend to have a more specialized customer base. Goldman Sachs is investment banking business, tends to focus more on corporations and customers that are more active in securities trading. Finally, we have the standalone investment banks like Lazard. These firms offer a full range of investment banking services but have a limited geographic footprint. They tend to just operate in major cities and they tend to focus on larger clients.

Corporate Structure

- Standalone versus affiliated
- Two waves
 - 1998 Gramm-Leach-Bliley Act
 - 2007-2009 financial crisis

I just set the term standalone without really explaining what it means. Well, this is an industry that has gone through some major changes. In the past, we had several prominent independent investment banks. JPMorgan, Merrill Lynch, Bear Stearns, Lehman Brothers, Morgan Stanley. Today the largest firms have merged with commercial banks and now sit within the holding company structure. This is what I mean by standalone versus affiliated with a commercial bank. Two merger waves brought us to this industry structure. First, regulatory changes in the late 1990s permitted commercial and investment banking activities to happen at the same firm. This cause several commercial banks to acquire or set up their own investment banks. And second, the financial crisis. This caused a wave of distressed acquisitions. By the end of 2008, Lehman Brothers was bankrupt, Bear Stearns and Merrill Lynch were acquired by commercial banks, and Goldman Sachs and Morgan Stanley had to reorganize as commercial banks in order to survive.

Merger / Acquisition	Year	Price (Billions)
Citicorp merges with Travelers	1998	\$83.0
JPMorgan Chase acquires Bank One	2004	\$60.0
Bank of America acquires Merrill Lynch	2008	\$50.0
Bank of America acquires FleetBoston	2003	\$49.3
Chase Manhattan merges with JPMorgan	2000	\$35.0
Bank of America acquires MBNA	2005	\$35.0
BB&T merges with SunTrust Banks	2019	\$28.0
Wachovia acquires Golden West Financial	2006	\$25.5
Wells Fargo acquires Wachovia	2008	\$15.1
Wachovia acquires Southtrust	2004	\$14.3

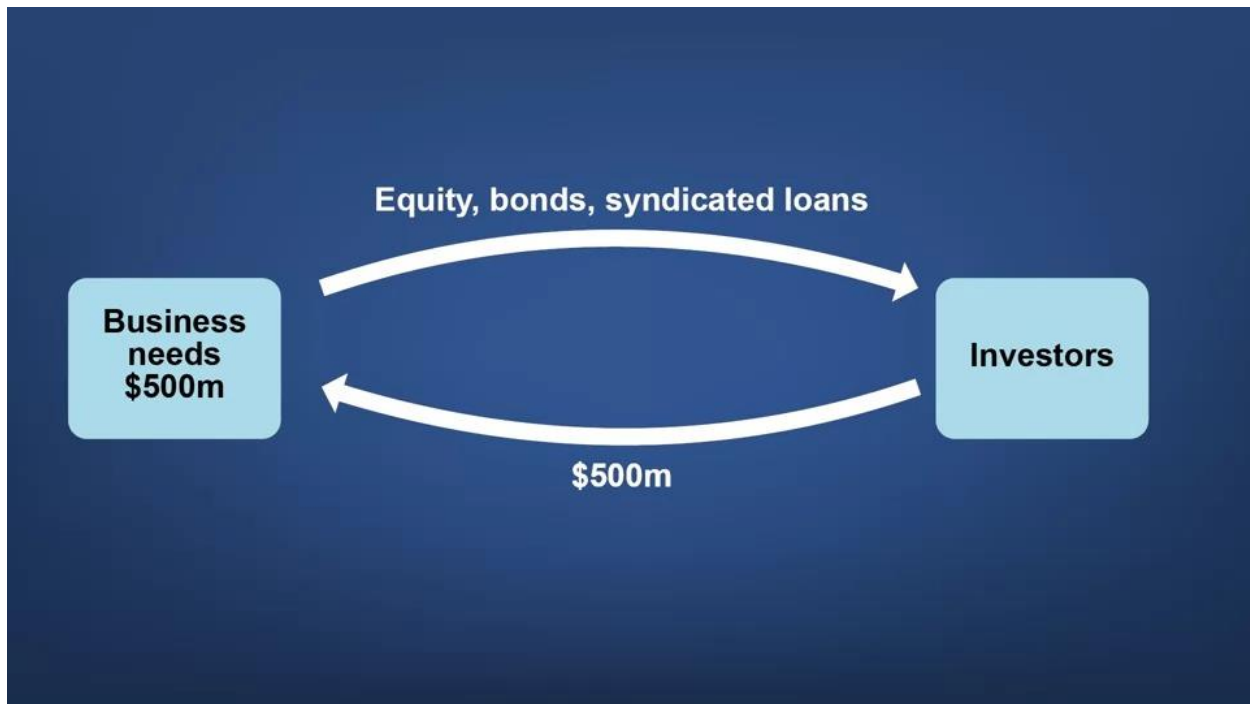
Here is a list of the 10 biggest combinations in the US investment banking industry. Citigroup was formed by the merger of Citicorp, a commercial bank and Travelers, which was an insurance company that owned an investment bank. This merger happened in 1998. At the same time as the passage of the Gramm-Leach-Bliley Act. Chase acquired JPMorgan soon after. Here is Bank of America acquiring Merrill Lynch in 2008. This was the largest distressed acquisition during that financial crisis.

Global Investment Banks FY 2020

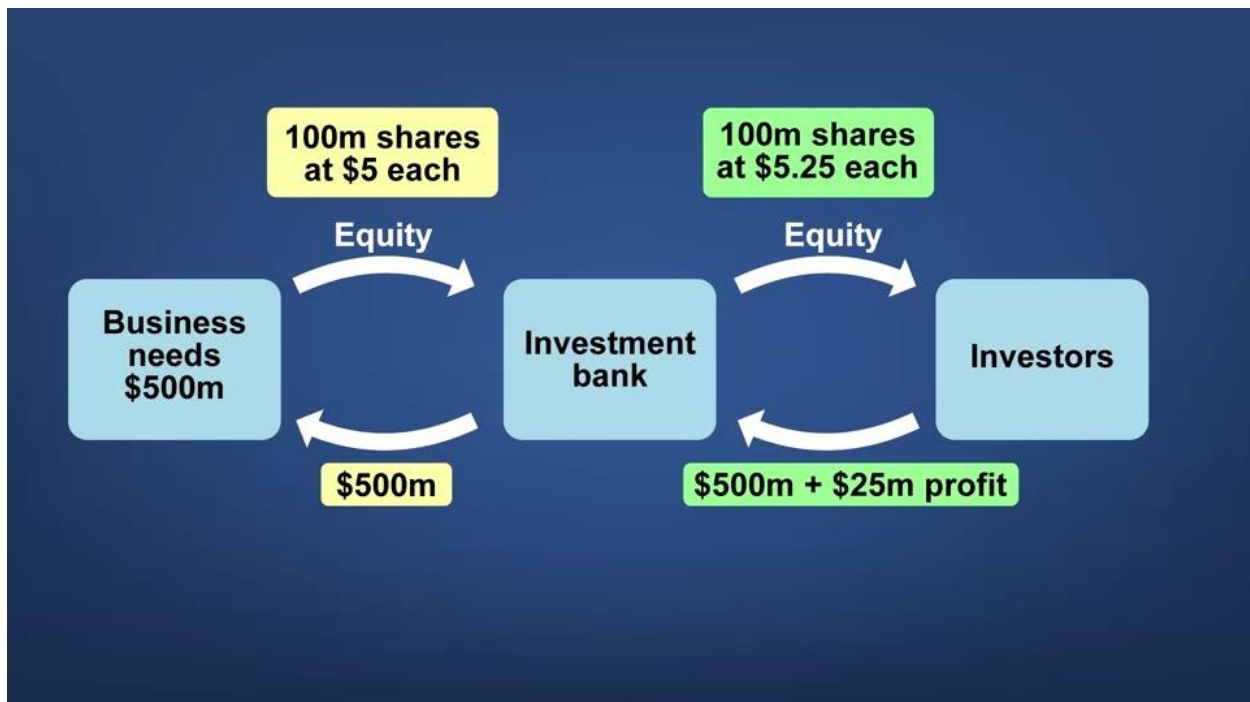
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2.3	Jefferies LLC	\$2,071.1
1.9	Wells Fargo Sec.	\$1,714.4
49.5	Subtotal	\$45,463.9
100.0	Total	\$91,847.7

Here is a breakdown of investment banking revenue in 2020. We can see the top ten investment banks globally. Only Jefferies is a stand-alone firm. The rest are affiliated with commercial banks and take deposits from retail customers. These 10 banks collect about 50 percent of the fees globally, and the market leader collects around 10 percent. So while there has been a lot of consolidation, this is still a fairly competitive market, at least on a global scale.

Lesson 2-2.5: Investment Banking Activities

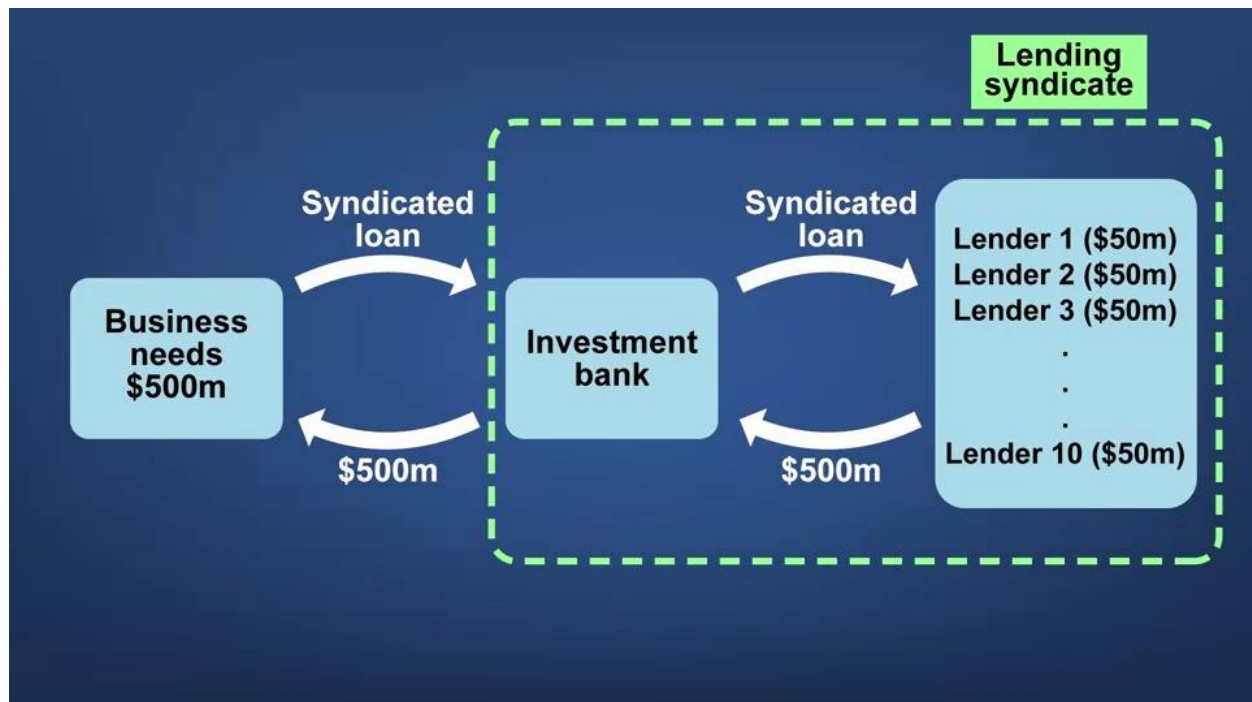


Ok, let's get into the nuts and bolts of the investment banking business model. First, let's take a closer look at securities underwriting. We will focus on equity, bond, and syndicated loan underwriting. This activity provides the bulk of investment banking fees. Let's say a company wants to raise \$500 million by selling financial instruments. This could be equity. It could be debt.



An investment bank will intermediate this process. In fact, usually several investment

banks will compete to win the mandate to lead that underwriting process. The lead underwriter will be responsible for setting the terms of the deal and managing the sale of the newly issued financial instruments. Let's consider the case of inequity sell. If a firm is selling equity for the first time, this is called an initial public offering or IPO. A listed firm can sell more equity in a follow-on deal. This is also called a seasoned offering. The investment bank will meet with the business and the potential investors to negotiate a price and interest in purchasing shares. This process is called book-building. The investment bank may be responsible for buying the entire issue at a guarantee price and selling it at another to outside investors. Hopefully, this price will be higher. The spread between the price they buy and the price that they sell determines their profits. Assuming the investment bank buys 100 million shares at a negotiated price of \$5 per share and sells those shares at a price of \$5.25 each, then a spread of \$0.25 will yield a nice profit of \$25 million. These profits are not guaranteed. Morgan Stanley famously mispriced Facebook's IPO and its share price lost nearly 50% in the subsequent weeks. It took a full 16 months to return back to the initial valuation of \$38 per share. The underwriter would like to set an offering price that clears its inventory at a profit. However, that price should reflect a fair valuation of the business. Otherwise the bank may risk damaging its reputation and future business prospects. Bond underwriting follows a similar process.



Let's instead consider a syndicated loan, that's a more interesting case. Usually, a commercial bank would ground a business line. However, when these loans are very big, this might be too risky. For these large loans, a group of banks and other lenders may come together and collectively fund the deal to our borrower. This allows them to spread the risk among themselves. Here, the underwriting **bang-boom** negotiate terms of the loan on behalf of the lending syndicate. They will be responsible for both the origination step and the servicing step in the lending process. Leading the syndicate will

generate a lot of fees for the underwriting bank. The other syndicate members who act more like silent partners that provide funding for the loan. These other lenders will earn the usual interest income from investing in a loan until it matures.

M&A Advisory

- Strategy
- Valuation
- Financing

Suppose Firm X would like to buy target firm Y or a segment of target firm Y but does not know the correct price. Suppose firm Z wants to diversify the division and needs help finding potential buyers. Investment banks are frequently involved in negotiating and executing deals of this kind. They conduct research on potential acquisition targets.. Or, they assist sellers to find buyers. They are experts in estimating the value of public and private companies using financial modeling techniques. And then in terms of financing, the M&A advising bank can also assist with underwriting new securities to finance the deal. Any of the underwriting options, equity, debt, syndicated loans. They're all on the table. Finalizing the details of M&A transactions can be complicated both from a financial and legal standpoint. Investment banks are well compensated for the advice and services that they provide.

Top Earners

Product	Bank	% Share	Revenue (Millions)
Equity Capital Markets	Goldman Sachs	11.2	\$26,472.7
Follow-On	Goldman Sachs	11.3	\$12,687.3
IPO	Goldman Sachs	10.1	\$10,866.5
Mergers & Acquisitions	Goldman Sachs	9.8	\$25,262.3
Debt Capital Markets	JPMorgan	8.8	\$29,624.8
Corporate Bond-Investment Grade	JPMorgan	8.0	\$16,223.4
Corporate Bond-High Yield	JPMorgan	11.8	\$6,993.6
Syndicated Lending	JPMorgan	11.1	\$10,488.9
Investment Grade	JPMorgan	13.6	\$1,404.8
Leveraged	JPMorgan	10.7	\$9,084.1

Ok, here are the various investment banking products. We have fees from underwriting equity, bonds, syndicated loans. Here we can see a breakdown by initial public offering and follow-on offerings. For debt, we see a breakdown by investment grade. These are companies with good credit ratings and then below investment grade firms. In 2020, total fee revenues from equity underwriting, from bond underwriting, and M&A related activities were all about the same order of magnitude. Fees from syndicated lending were about half as much.

Securities Trading

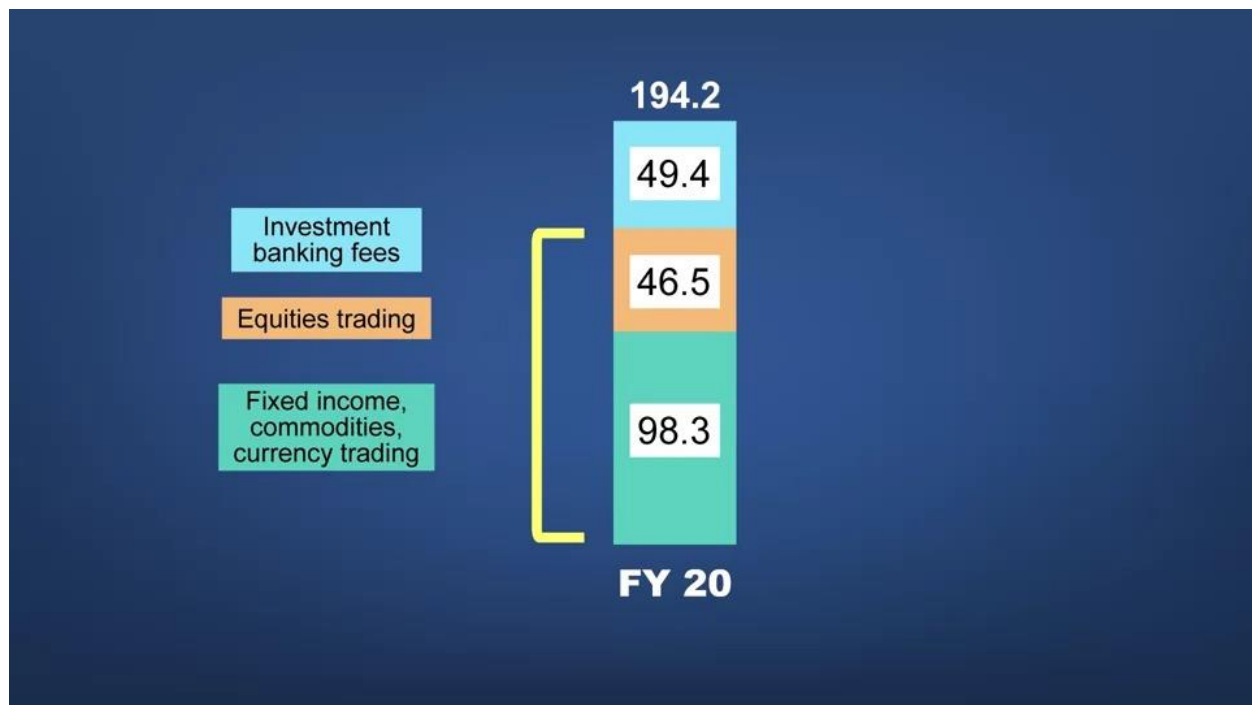
Market making

- Broker
- Dealer

Proprietary trading

- Position
- Arbitrage

Securities trading is the other key activity of investment banks. This consists of market-making and proprietary trading. Market-making occurs when an investment bank creates a market where investors can come together and trade securities. There are many other securities firms that do this, but investment banks play a central role. Brokers execute trades on behalf of their customers, earning a fee or commission along the way. Let's say our investment bank buys a Facebook stock at an ask price of \$20 and sells it to a customer at the bid price of \$21. Then it will earn \$1 profit. This profit is called the bid-ask spread. The dealer arm of an investment bank may carry some stock in his inventory. In this case you might find yourself buying the Facebook stock from the bank itself rather than the bank trading in markets on your behalf. Combining securities dealing and underwriting has clear synergies. If I'm an investment bank that underwrites a new stock issue then it is easy for me to carry some leftover stock in my inventory. Proprietary trading is when a bank takes an active position in a financial instrument to make a profit. They are not assisting customers, they are trying to make a profit for themselves. A position trade involves taking a long or short position in a security for several weeks or several months in anticipation of a favorable price movements. Arbitrage trading usually involves exploiting small temporary mispricing in financial markets. This mispricing should not exist, at least not in theory, and so this presents an opportunity to make a profit. It can also make markets more liquid and prices more efficient.



Here is one final chart. It shows total investment banking and trading revenues for the 12 largest U.S. and European institutions. Total revenues in 2020 were just shy of \$200 billion. About \$150 billion of that came from fixed income and equity trading. So that's around 75%. The remaining 25 percent came from investment banking fees.

Summary

- Business model
 - Underwriting, trading, M&A
- Corporate structure

Ok, in this session, we have covered the main activities of investment banks. Securities underwriting. Investment banks help firms issue new securities and place them with investors. Trading. Investment banks assist in the trading of securities that are already

out there and then there's corporate strategy. All three avenues are important sources of fee-based income for investment banks. Finally, this is an industry that has gone through some dramatic changes.

Standalone investment banks are few and far between. The largest investment banks now sit within bank holding companies structures.

Lesson 2-3: Financial Statement Analysis of Modern Banks

[Lesson 2-3.1: Balance Sheet](#)

Learning Objectives

- Balance sheet
 - $\text{Assets} = \text{liabilities} + \text{equity}$
- JP Morgan Chase
- All banks

We have discussed how banks are organized as firms and we have discussed how the banking industry and bank business model has evolved. We know that commercial banks mostly raise deposits and grant loans. Some of the larger firms also do investment banking. In this session, we will firm up our understanding of what banks do. We will take a close look at the balance sheets of commercial bank holding companies. Remember that a balance sheet provides a list of investments on the asset side and sources of funds on the liability side. We will examine JP Morgan Chase's balance sheet as a case study. We will also examine what is called the consolidated balance sheet of the entire commercial banking industry.

Assets	Liabilities
Cash	Deposits
Loans	Other borrowed funds
Premises and equipment	Capital
Other assets	

Here is a simple representation of the balance sheet of a commercial bank. On the asset side of the balance sheet, we can see what banks do with the funds they raise. Some cash in the vault, we have loans. These could be mortgages, business loans, or loans to households. There are the tables and chairs that the loan officers sit on, and there are some other financial assets. These can be government bonds, corporate bonds, and other securities. On the liability side, we see how the bank is funded. This will include some customer deposits, checking and savings accounts. The bank may also raise some wholesale funding in the money market. Bank owners may also retain some earnings within the firm rather than paying them out, or sell some additional shares. The larger commercial bank holding companies have significant market making and securities trading activities.

Banking Book	Assets Cash Loans Premises and equipment Other assets	Liabilities Deposits Other borrowed funds Capital
Trading Book	Assets Bonds (long) Commodities (long) FX (long) Equities (long) Mortgage-backed securities (long)	Liabilities Bonds (short) Commodities (short) FX (short) Equities (short)

We, therefore, separate out the balance sheet into the banking book and the trading book. The banking book represents the long-term borrowing and lending activities of our traditional bank. These activities produce some interest income and fees from lending. The trading book contains investments in securities, commodities, and currencies that can be quickly bought and sold in financial markets. Long positions are recorded at market value on the asset side; short positions which involve borrowing are recorded as liabilities. The trading portfolio produces trading gains and losses that are booked as non-interest income and expenses. We shall see where these profits show up in the income statement in another session.

Balance Sheet - JPM

December 31, (in millions, except share data)	2020
Assets	
Cash and due from banks	\$ 24,874
Deposits with banks	502,735
Federal funds sold and securities purchased under resale agreements (included \$238,015 and \$14,561 at fair value)	296,284
Securities borrowed (included \$52,983 and \$6,237 at fair value)	160,635
Trading assets (included assets pledged of \$130,645 and \$111,522) ⁽¹⁾	503,126
Available-for-sale securities (amortized cost of \$381,729 and \$345,306; included assets pledged of \$32,227 and \$10,325)	388,178
Held-to-maturity securities (net of allowance for credit losses of \$78)	201,821
Investment securities, net of allowance for credit losses	589,999
Loans (included \$44,474 and \$44,955 at fair value) ⁽²⁾	1,012,853
Allowance for loan losses	(28,328)
Loans, net of allowance for loan losses	984,525
Accrued interest and accounts receivable	90,503
Premises and equipment	27,109
Goodwill, MSRs and other intangible assets	53,428
Other assets (included \$13,827 and \$12,676 at fair value and assets pledged of \$3,739 and \$3,349) ⁽³⁾	152,853
Total assets⁽⁴⁾	\$ 3,386,071
Liabilities	
Deposits (included \$14,484 and \$28,589 at fair value)	\$ 2,144,257
Federal funds purchased and securities loaned or sold under repurchase agreements (included \$155,735 and \$549 at fair value)	215,209
Short-term borrowings (included \$16,893 and \$5,920 at fair value)	45,208
Trading liabilities	170,181
Accounts payable and other liabilities (included \$3,476 and \$3,728 at fair value)	232,599
Beneficial interests issued by consolidated VIEs (included \$41 and \$36 at fair value)	17,578
Long-term debt (included \$76,817 and \$75,745 at fair value)	281,685
Total liabilities⁽⁵⁾	3,106,717
Commitments and contingencies (refer to Notes 28, 29 and 30)	
Stockholders' equity	
Preferred stock (\$1 par value; authorized 200,000,000 shares; issued 3,006,250 and 2,699,250 shares)	30,063
Common stock (\$1 par value; authorized 9,000,000,000 shares; issued 4,104,933,895 shares)	4,105
Additional paid-in capital	88,394
Retained earnings	236,990
Accumulated other comprehensive income	7,986
Shares held in restricted stock units ("RSUs") trust, at cost (zero and 472,953 shares)	—
Treasury stock, at cost (1,055,499,435 and 1,020,912,567 shares)	(88,184)
Total stockholders' equity	279,354
Total liabilities and stockholders' equity	\$ 3,386,071

Let's take a look at JP Morgan Chase's balance sheet. Please keep in mind that this is one of the largest and most complex banks on the planet. I'm going to focus on the most important items, and even then this is going to be a bit messy. Okay. Here we go. This is the consolidated balance sheet across the retail and wholesale business segments, 2020 data, and all the numbers you see are in millions of dollars. We see assets. Total assets are about \$3.4 trillion. Yes, that's trillion with a T. We see liabilities. One way or another this is borrowed money. Liabilities add up to about \$3.1 trillion. Total funding – that's debt plus equity – has to equal assets. So stockholder's equity must be equal to the difference between assets and liabilities, so about \$300 billion.

Assets - JPM

December 31, (in millions, except share data)	2020
Assets	
Cash and due from banks	\$ 24,874
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Other assets (included \$13,827 and \$12,676 at fair value and assets pledged of \$3,739 and \$3,349) ^(a)	152,853
Total assets^(a)	\$ 3,386,071

Let's take a closer look at the asset side. Which investments are most important for JP Morgan? First, we can see cash held at the bank plus cash parked in deposit accounts at other banks. These deposit accounts at other banks help facilitate payments between banks and they might also earn some interest. These two items are usually added up in cold cash. Cash is the most liquid asset that banks carry in their balance sheets. Second, we have federal funds sold and reverse repo agreements. This is JP Morgan lending to other financial institutions, such as banks on a short-term basis. Third, we have trading account assets. These are securities held in a short-term basis with the intent to sell as part of a position or arbitrage trade. Investment securities, on the other hand, are long-term investments in debt securities that are not part of the bank's trading portfolio. Finally, we have loans. This is the largest item in the balance sheet, about one trillion dollars altogether. We can take a closer look at the loans in the retail arm of JP Morgan's business.

Retail Business - JPM

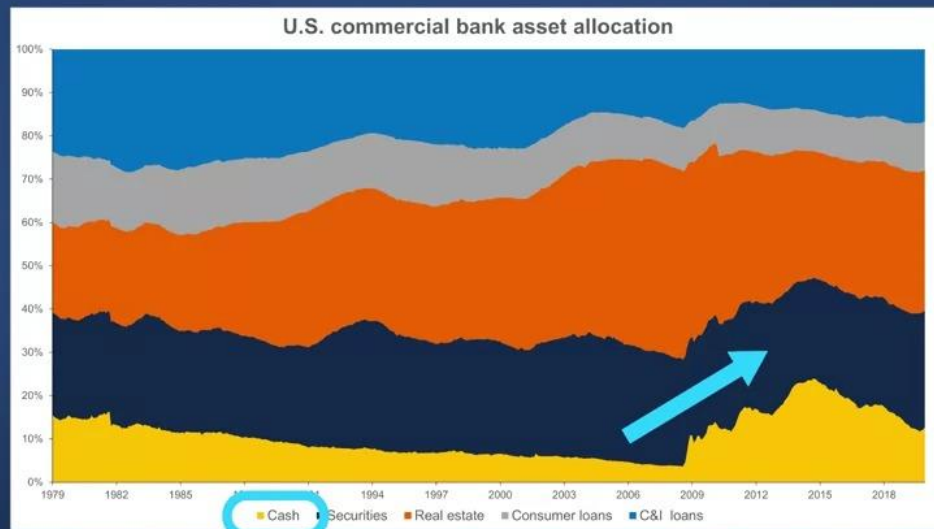
As of or for the year ended December 31, (in millions, except headcount)		2020
Selected balance sheet data (period-end)		
Total assets		\$ 496,705
Loans:		
Consumer & Business Banking		48,810
Home Lending ^{(a)(b)}		182,121
Card		144,216
Auto		66,432
Total loans		441,579
Deposits		958,706

Of the roughly one trillion dollars of loans, about half are assets in the retail arm. The other \$500 billion of loans must be coming from the wholesale arm. So that's large loans to corporations and governments. We have loans to individuals and small businesses, loans that are backed by residential real estate, credit cards, and also auto loans. You might have spotted this, but it looks like JP Morgan through its quite extensive branch network raises about one trillion dollars of deposits. This is more than the retail banking business is lending out. So JP Morgan and other banks do the same thing, use relatively cheap and sticky deposit funding to support the other investments that it makes.

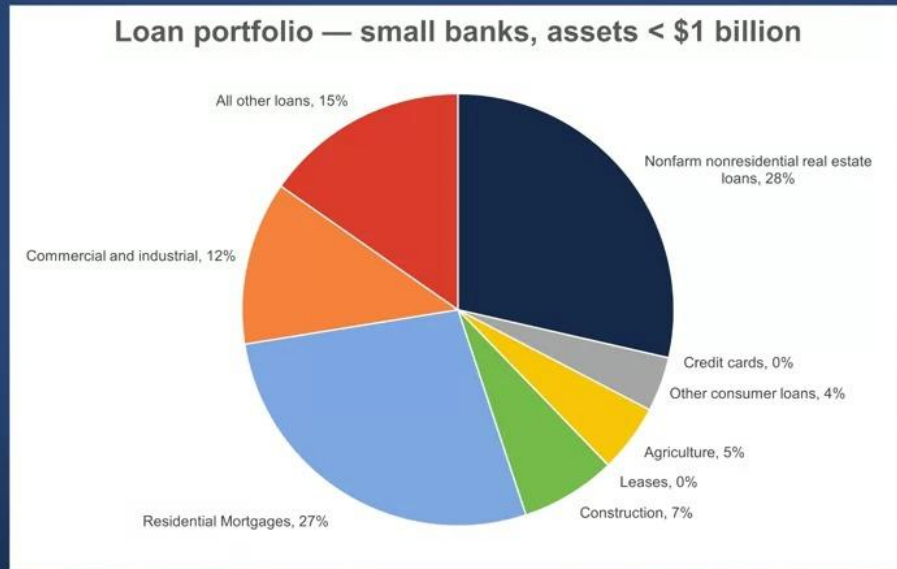
Assets (\$bn) – All US Banks in 2018

Loans and Securities		\$13,445.4
Investment Securities		\$4,302.5
US Government Securities	\$2,400.5	
Other	\$1,902.0	
Total Loans		\$9,142.9
Interbank Loans	\$80.8	
Loans Excluding Interbank	\$9,062.1	
Commercial and Industrial	\$2,038.7	
Real Estate	\$4,390.4	
Individual	\$1,549.1	
All Other	\$1,198.9	
Less: Reserve for Loan Losses	\$115.0	
Total Cash Assets		\$1,760.6
Other Assets		\$1,274.4
Total Assets		\$16,480.4

Now, let's repeat this exercise for the entire banking industry as a whole. Here is what's called the consolidated balance sheets of all U.S. commercial banks. This is for the year 2018. It covers roughly 5,000 commercial banks that existed in the US at that time. The government collects these statistics as they monitor the safety and soundness of the banking system. Loans are about nine trillion dollars or roughly 55% of the \$16.5 trillion of total assets. Commercial and residential real estate loans, business loans, and then loans to individuals. These are the three largest categories of loans in that order of importance. Investment securities are the second most important category with about 25% of assets. U.S. government securities such as treasury bonds and agency mortgage bank securities are the largest subcomponents. These assets can be easily sold and are a good backup for a bank's cash reserves. Cash itself is roughly 10% of assets.

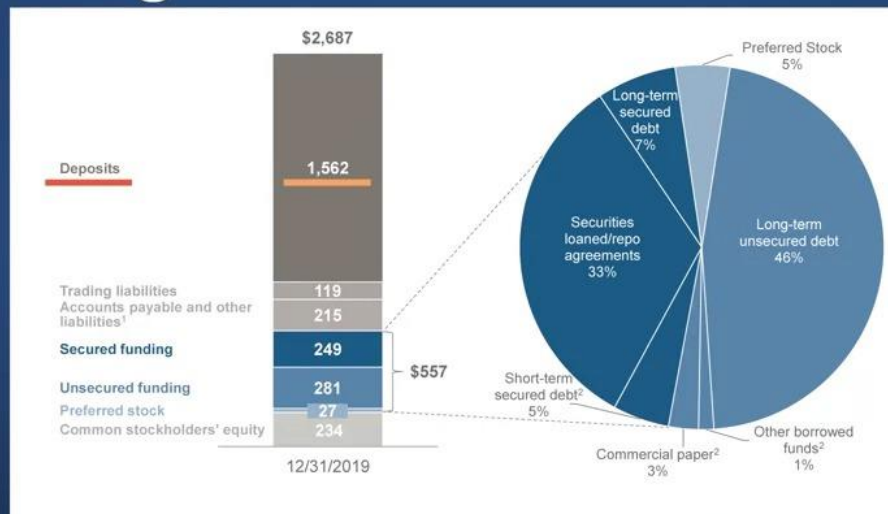


This chart shows the main investments of banks, securities, real estate, consumer credits, business loans, and cash. Cash holdings among banks are actually near historic highs. If we look back at the past 40 years or so, banks usually prefer to hold as little cash as possible. Every dollar that sits idle in the vault could be invested. However, there was a financial crash in 2008, at which time many banks ran out of cash. And in the years since 2008, interest rates have been historically low. This has lowered the opportunity cost of holding cash. So far, we have considered the assets of the biggest bank in the U.S., JP Morgan and an average bank. What about the smallest community banks?



This chart shows the loan portfolio allocation for banks with less than one billion dollars of total assets. These are all local retail banks. Here are a couple of things worth noting. First, they operate on such a small scale that credit cards do not play any role in their business. On a per dollar basis, these loans are just too expensive to originate in service. Instead, bigger banks and specialized finance companies deal with large pools of credit cards. Second, business lending plays a smaller role than you might have thought. Larger banks tend to take care of most of the commercial lending as well. What the community banks do? They focus on real estate lending. This includes residential and commercial real estate, as well as construction and development loans. They know their local markets very well and this seems to be their comparative advantage.

Funding - JPM



Now let's move on to the funding side of the balance sheet. Those investments are either funded by debt or by equity. Commercial banks get funds from depositors and they can borrow in money markets. This is debt. Banks can also use equity provided by their owners. Here is JP Morgan Chase. Deposits are by far the largest source of funding for this bank. This includes checking accounts which are also called transaction accounts. The deposits bucket also includes non-transaction accounts. These are mostly certificates of deposit which look a lot like short-term bonds. The second most important source of funds is borrowed money. At one extreme, this includes short-term secure debts. For JP Morgan, this means borrowing through the repo market and at the other extreme we have long-term unsecured debt. In this case, JP Morgan issues a large amount of long-dated senior bonds. The other categories of debt are less important. The last source of funding is bank equity capital. This is funding that comes from the owners of the bank.

Equity Funding - JPM

December 31, (in millions, except share data)	2020
Stockholders' equity	
Preferred stock (\$1 par value; authorized 200,000,000 shares; issued 3,006,250 and 2,699,250 shares)	30,063
Common stock (\$1 par value; authorized 9,000,000,000 shares; issued 4,104,933,895 shares)	4,105
Additional paid-in capital	88,394
Retained earnings	236,990
Accumulated other comprehensive income	7,986
Shares held in restricted stock units ("RSU") trust, at cost (zero and 472,953 shares)	—
Treasury stock, at cost (1,055,499,435 and 1,020,912,567 shares)	(88,184)
Total stockholders' equity	279,354
Total liabilities and stockholders' equity	\$ 3,386,071

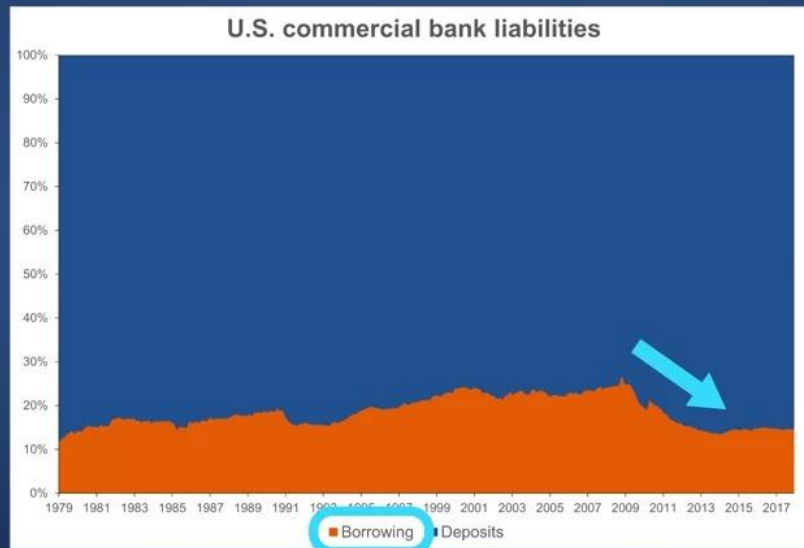
This consists of paid in capital from the issuance of common and preferred stock. However, the majority of JP Morgan's equity comes from retained earnings that are not paid out. These are profits that are reinvested into the business. One final point. You might have noticed that JP Morgan uses a lot of debt to finance investments. This is a common feature of banks and JP Morgan is not an exception. The share of equity and total funding is around eight percent. Put differently, the bank's equity multiplier is around 12 times.

Funding (\$bn) – All US Banks in 2018

A = \$16,480

Total Deposits			\$12,631.6
Deposits Held in Foreign Offices	\$1,251.8		
Deposits Held in Domestic Offices	\$11,379.8		77%
Transactions Accounts	\$1,969.4		
Nontransaction Accounts	\$9,410.4		
Borrowings		\$1,589.4	
Other Liabilities		\$390.3	
Total Liabilities		\$14,611.3	
Total Equity Capital		\$1,869.1	

Here are the same sources of funds but now for the banking industry as a whole. Again, this is 2018 data. The main sources of funds are deposits, borrowings, and equity capital. Deposits are mostly held in domestic offices in the United States as opposed to overseas. Most of these banks only have domestic operations after all. As we saw earlier, total assets for the banking industry in 2018 was about \$16.5 trillion. Let's make a note of that. Then we can see deposits fund about 77 percent of the investments that banks make. Non-deposit borrowings are about 11% of assets, and equity capital which is total assets minus liabilities funds the remaining 11% of assets. There is some rounding error here, so these percentages don't quite add up to 100.



When we look at the liabilities of banks over a long time horizon, we can see just how important deposits are. This is data from the 1970s to present. Deposits have been steady between 70% and 80% of total commercial bank assets. Since the 2008 financial crisis, banks have become more cautious and our sticky and reliable deposits now play an even more important role.

Summary

- Balance sheets
- Recent trends

Let's wrap up. In this session, we have analyzed the balance sheet of JP Morgan and the banking industry as a whole. We have seen data on the key sources and uses of

funds at these banks. Loans, securities, and cash on the asset side. Deposits, borrowings and capital markets, and equity capital on the funding side. Since the 2008 financial crisis, banks are carrying more cash and using more deposits to fund their investments. Many banks failed at that time. So nowadays, banks are behaving much more cautiously.

Lesson 2-3.2: Off Balance Sheet

Learning Objectives

- Contingent exposures
- Common OBS activities

In this session, we are going to discuss the off-balance sheet activities of banks. We have two goals. First, I want to introduce the concept of a contingent asset and a contingent liability. These are items that do not show up in the balance sheets that we have seen so far. These items do not show up in the banking book or the trading portfolio of a modern bank. However, on the current accounting standards, banks must record this information in their financial statements. This ensures that investors and regulators are well-informed. Second, I want to describe the most important off-balance sheet or OBS activities at banks. I want to give you a sense of the size and the scale of this stuff.

Off Balance Sheet

- Why? Fee income
- OBS activities
 - Contingency
 - Assets versus liabilities

Okay. Let's get right to it. To generate fee-based income, banks engage in a broad array of activities off the balance sheet. Some of these activities resembled traditional bank lending. Other items look more like insurance. What's important to remember is that off-balance-sheet activities are contingent in nature. This means that when an event occurs, the off-balance sheet item has a material impact on the banks' balance sheets or cash flows. An off-balance sheet assets moves onto the asset side of the balance sheet or affects the bank's income when a pre-specified event occurs. The simplest example to think of here is a credit card. We are all familiar with credit cards they have credit limits that tell us the maximum amount we can use. A bank has provided a commitment to lend as much as that limit. Bank's record undrawn credit card commitments as off-balance sheet assets. When I charge my credit card, this is the contingent events, me charging the card, the bank will convert an off-balance sheet assets into an on-balance sheet assets, a consumer loan. In the same way, off-balance sheet liabilities will show up on the liability side of the balance sheet or affect the banks expenses when an event occurs. These items haven't shown up on the balance sheet yet, but they might in the future. Guarantees, which act as a form of insurance are common here. The bank writes the insurance today and records it the off the balance sheet. If there is an insurance claim tomorrow because the pre-specified event happens, then this is going to have a material impact on the banks cash flows. There will be a payout. Why do you banks like issuing credit cards and writing insurance? Well, because it generates upfront fees.

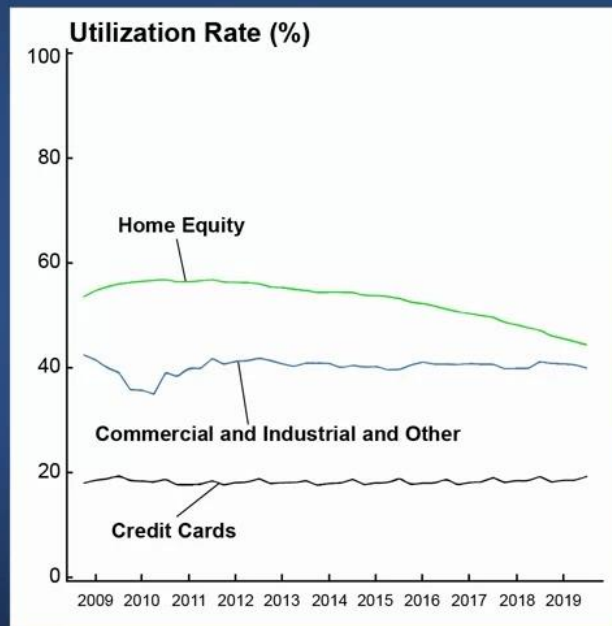
Balance Sheet	Banking Book	Assets Cash Loans Premises and equipment Other assets	Liabilities Deposits Other borrowed funds Capital
	Trading Book	Bonds (long) Commodities (long) FX (long) Equities (long) Mortgage-backed securities (long)	Bonds (short) Commodities (short) FX (short) Equities (short)
	Off Balance Sheet	Loan commitments Derivatives (long)	Guarantees Derivatives (short)

Here's the balance sheet. We have discussed the bank book and the trading book previously and now here is the off-balance sheets. The most important items on the asset side are undrawn loan commitments and long derivatives positions. Credit cards can be charged in the future and this will create loans in the banking book. Long derivatives positions can make profits or losses in the future. What about contingent liabilities? Here we also have derivatives, but now these are considered short positions that can also make profits or losses and we have our guarantees. These guarantees can take a variety of forms but they mostly behave like insurance. Under some conditions, the bank will have to make an insurance payouts.

OBS Activities - All U.S. Banks

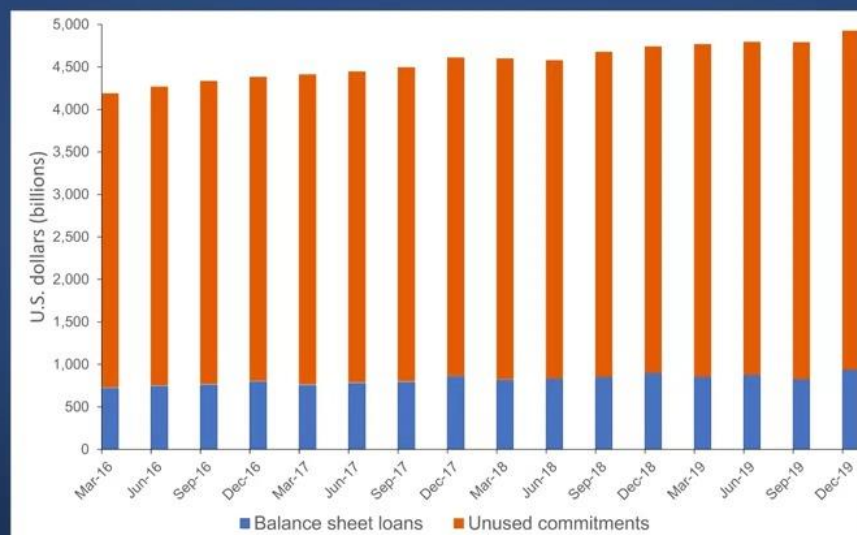
(\$ billions)	1992	2018
➡ Unused Loan Commitments	\$1,281	\$7,722
Notional Amount of Derivatives:		
Interest Rate Derivatives	\$4,872	\$155,469
Foreign Exchange Derivatives	3,789	38,839
Equity & Commodity Derivatives	102	5,098
Credit Derivatives	—	4,345
➡ Total Notional Amount of Derivatives	8,763	203,751
Loan Commitments + Derivatives	10,044	211,473
Total Assets (on-balance-sheet items)	3,506	17,531

Here is the data on the key off-balance sheet activities for all commercial banks in the United States. We have 1992, which is ancient history, and then we have the 2018 data. This information is collected by government agencies that want to make sure that banks aren't taking too many risks in these business lines. We are going to pay special attention to the unused loan commitments and derivatives. These are by far the most important activities. Let's start with loan commitments. A loan commitment is a legally binding promise to provide funds up to a certain maximum limits and I'd say pre-specified interest rate. These promises can be made to households or businesses. They can be unsecured credit cards or lines of credit secured by houses. Banks must stand by and be ready to provide funds at any time within the commitment period. And when that card is charged a loan will arrive in the bank's balance sheet.



This chart is from the FDIC. It shows the utilization rates of the various types of loan commitments in recent years – home equity loans, business loans, credit cards. Less than 100 % means that there is some undrawn amount sitting off the balance sheet somewhere.

Credit Cards - All U.S. Banks



Let's zoom in on the credit cards. Here we can see some dollar amounts for both the drawn and undrawn credit cards. At no point in time did the drawn balances exceed \$1 trillion. However, we can see that 4 to 5 times that amount is undrawn. Banks can earn

fees on these undrawn balances. Regulators require that banks disclose these undrawn balances in their financial statements. Why? Because excessive drawdowns can create risks at banks. Now let's talk about derivatives. Banks can use derivatives to reduce their own risk exposure to say unfavorable changes in interest rates. Banks can also act as dealers and take the other side of a derivatives trader with a customer for a fee.

OBS Activities - All U.S. Banks

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In either case, the derivatives that are bought and sold may lead to some cash inflow or a cash outflow in the future. The bank therefore must record these transactions off the balance sheet. The reporting here is done in notional amounts, so for interest rate derivatives, which is by far the biggest item. This is a jaw-dropping number, \$150 trillion. If we think of derivatives as providing insurance, that's how I like to think of it, then this tells us about the total value of items being insured.

OBS Activities - All U.S. Banks

(\$ billions)	1992	2018
Unused Loan Commitments	\$1,281	\$7,722 → 5x
Notional Amount of Derivatives:		
Interest Rate Derivatives	\$4,872	\$155,469
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Equity & Commodity Derivatives	102	5,098
Credit Derivatives	—	4,345
Total Notional Amount of Derivatives	8,763	203,751
Loan Commitments + Derivatives	10,044	211,473
Total Assets (on-balance-sheet items)	3,506	17,531 → 5x

However, this large number doesn't tell us much about the potential cashflow exposure that the bank has or the market value of these positions. For example, I might have \$1 millions of home insurance. This is the notional amount but the insurance costs me \$2,000. That's the market value of the insurance contract. While notional amounts don't mean much the growth rates are actually meaningful. Loan commitments have grown more or less in line with the bank balance sheet in the past 30 years about a 5x increase.

OBS Activities - All U.S. Banks

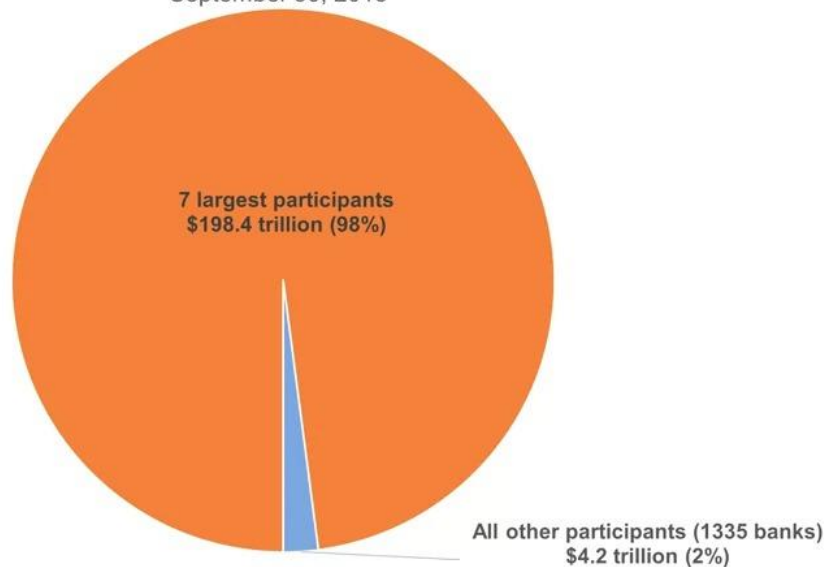
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Total Notional Amount of Derivatives	8,763	203,751
Loan Commitments + Derivatives	10,044	211,473
Total Assets (on-balance-sheet items)	3,506	17,531

→ 20x

Derivatives have shown explosive growth during this time period. Growth has been by a factor of 20.

Market concentration of derivatives

Notional amounts
September 30, 2018



Much of this derivatives activity is actually concentrated among the largest financial institutions: JP Morgan Chase, Citi Bank, Bank of America, Goldman Sachs, Morgan Stanley, Wells Fargo, HSBC. These are the largest dealer banks and having all of this activity concentrated in a handful of institutions, reflects economies of scale in this business.

Derivatives - JPM

Notional amount of derivative contracts
The following table summarizes the notional amount of derivative contracts outstanding as of December 31, 2020 and 2019.

December 31, (in billions)	Notional amounts ⁽¹⁾	
	2020	2019
Interest rate contracts		
Swaps	\$ 20,986	\$ 21,228
Futures and forwards	3,057	3,152
Written options	3,375	3,938
Purchased options	3,675	4,361
Total interest rate contracts	31,093	32,679
Credit derivatives⁽²⁾	1,201	1,242
Foreign exchange contracts		
Cross-currency swaps	3,924	3,604
Spot, futures and forwards	6,871	5,577
Written options	830	700
Purchased options	825	718
Total foreign exchange contracts	12,450	10,599
Equity contracts		
Swaps	448	406
Futures and forwards	140	142
Written options	676	646
Purchased options	621	611
Total equity contracts	1,885	1,805
Commodity contracts		
Swaps	138	147
Spot, futures and forwards	198	211
Written options	124	135
Purchased options	105	124
Total commodity contracts	565	617
Total derivative notional amounts	\$ 47,194	\$ 46,942

Let me illustrate this point with JP Morgan Chase. Here is JP Morgan's derivatives exposure in 2020: interest rate derivatives, credit derivatives, FX, equity, and commodity contracts. This activity is shown in notional amounts as we've seen so far. What has been missing from our discussion is that a mega bank like JP Morgan is taking positions in thousands of derivatives transactions. Most of this activities on behalf of customers. Here if you look at the interest rate options, you can see JP Morgan is both purchasing and selling or writing options at the same time. This may naturally lead to some canceling out or netting of exposures. Larger banks with more and more transactions may naturally see more netting. You can see this quite clearly with JP Morgan's credit default swaps book. You can think of a credit default swap or CDS as an insurance contract on a bond or a loan. If the credit defaults, that's our contingency, then the insurance pays out. What do the numbers show us?

Credit Derivatives - JPM

December 31, 2020 (in millions)	Maximum payout/Notional amount			
	Protection sold	Protection purchased with identical underlyings ⁽¹⁾	Net protection (sold)/purchased ⁽²⁾	Other protection purchased ⁽³⁾
Credit derivatives				
Credit default swaps	\$ (535,094)	\$ 554,565	\$ 19,471	\$ 4,001
Other credit derivatives ⁽⁴⁾	(40,084)	57,344	17,260	9,415
Total credit derivatives	(575,178)	611,909	36,731	13,416
Credit-related notes	—	—	—	10,248
Total	\$ (575,178)	\$ 611,909	\$ 36,731	\$ 23,664

Well, JP Morgan is involved in about \$1 trillion worth of CDS, mind-boggling sums. However, this is spread out equally between insurance sold and insurance bought. Overall, the net exposure is much smaller. This is how netting works. Hopefully, JP Morgan pays a lot less for the insurance that it buys than the fees it collects from selling insurance. Then this activity will make a nice profits without creating too much risk exposure for the bank. The off-balance sheet contains a list of contingent exposures. They have potential to impact the bank's cash flows at some point in the future.

Summary

- Contingent exposures
 - Loan commitments
 - Derivatives
- Fee income

We saw that loan commitments and derivatives are the most important off-balance-sheet activities. They generate the most fee income for the bank. We've said nothing about risk in this session. Risks that were buried off the balance sheet were at the heart of the 2008 financial crisis. These poorly understood risks led to the failure, distressed acquisition, or bailout of some of the largest financial institutions in the world. This led to a barrage of new regulations. A very interesting topic for another time.

Lesson 2-3.3: Income Statement

Learning Objectives

- Income statement
- Interest income versus fees

Commercial banks accept deposits and use the proceeds to make loans to individuals, loans to businesses and real estate loans. Banks borrow, lend, and earn the spread. The spread is the difference between interest income earned on lending and the interest expense paid on deposits. Is this the whole story? Well, not quite. It's a little bit more complicated than that. We have seen that modern banks are large and diversified businesses. They earn interest income from their lending business and fee revenue from other business lines. In this session, we are going to analyze the income statement of a bank. We are going to continue with JP Morgan Chase as a case study. We will see how the various business lines of JP Morgan generate both interest income and expenses as well as fee revenue. This will tell us exactly how a modern bank turns a profit.

Income Statement

- Net interest income
 - Interest income
 - Interest expense
- Noninterest income (fees)
- Noninterest expense

We can break down the income statement of a bank into three parts. The first tells us about the profitability of the lending business. This is the traditional bank. It grants loans. It invests in securities. This generates interest income. It raises funds from depositors and borrows in financial markets in order to fund those investments. That borrowing doesn't come for free. Banks have to pay some interest expense. Net interest income is equal to interest income minus interest expense. This is the top line measure of bank profitability. What's left? Two items. First, we know that modern banks generate a lot of fee income. This is called non-interest revenue. Second is what's called non-interest expense. This is a catch-all for any expense that is unrelated to borrowing. It includes everything from tables and chairs to wages and bonuses. Let's study all of these objects in the context of JP Morgan Chase.

JPMorgan Chase						
Consumer Businesses			Wholesale Businesses			
Consumer & Community Banking			Corporate & Investment Bank		Commercial Banking	Asset & Wealth Management
Consumer & Business Banking	Home Lending	Card & Auto	Banking	Markets & Securities Services		
- Consumer Banking - J.P. Morgan Wealth Management - Business Banking	- Home Lending Production - Home Lending Servicing - Real Estate Portfolios	- Credit Card - Auto	- Investment Banking - Wholesale Payments - Lending	- Fixed Income Markets - Equity Markets - Securities Services - Credit Adjustments & Other	- Middle Market Banking - Corporate Client Banking - Commercial Real Estate Banking	- Asset Management - Wealth Management

As we've seen before, JP Morgan's business can be broken down into two big parts. First is the traditional bank that does the lending. This contains the retail banking arm as well as the commercial lending arm within the wholesale business. This is where our net interest income is coming from. Second, we're going to see non-interest income. Remember, that's fee income. It's mostly coming from the investment banking arm. This is mainly the underwriting business, market making, and asset management.

Net Interest Income - JPM

Year ended December 31, (in millions)	2020
Interest income	
Loans	\$ 43,758
Taxable securities	7,843
Non-taxable securities	1,184
Total investment securities	9,027
Trading assets - debt instruments	7,832
Federal funds sold and securities purchased under resale agreements	2,436
Securities borrowed	(302)
Deposits with banks	749
All other interest-earning assets	1,023
Total interest income	\$ 64,523
Interest expense	
Interest bearing deposits	\$ 2,357
Federal funds purchased and securities loaned or sold under repurchase agreements	1,058
Short-term borrowings	372
Trading liabilities - debt and all other interest-bearing liabilities	195
Long-term debt	5,764
Beneficial interest issued by consolidated VIEs	214
Total interest expense	\$ 9,960
Net interest income	\$ 54,563
Provision for credit losses	17,480
Net interest income after provision for credit losses	\$ 37,083

Here is the interest income and interest expense for JP Morgan. We have interest

income from the loan portfolio, from investment securities held in the banking book and trading portfolio, interest on loans made to other financial institutions. Add that all up and we arrive at the total interest income for the bank which was about \$65 billion in 2020. Interest expense is paid on deposits. It is also paid on other borrowings, these are both short-term and long-term debt. This will include coupon payments on long-term bonds that JP Morgan issues. Sum across all these items and we have our interest expense. Interest income, interest expense. The difference between the two is our net interest income about \$55 billion in 2020. This is the profitability of JP Morgan's lending business.

Net Income - JPM

Year ended December 31, (in millions, except per share data)	2020
Revenue	
Investment banking fees	\$ 9,486
Principal transactions	18,021
Lending- and deposit-related fees ⁽¹⁾	6,511
Asset management, administration and commissions ⁽²⁾	18,177
Investment securities gains/(losses)	802
Mortgage fees and related income	3,091
Card income ⁽³⁾	4,435
Other income	4,457
Noninterest revenue	64,980
Interest income	64,523
Interest expense	9,960
Net interest income	54,563
Total net revenue	119,543
Provision for credit losses	17,480
Noninterest expense	
Compensation expense	34,988
Occupancy expense	4,449
Technology, communications and equipment expense	10,338
Professional and outside services	8,464
Marketing ⁽⁴⁾	2,476
Other expense	5,941
Total noninterest expense	66,656
Income before income tax expense	35,407
Income tax expense	6,276
Net income	\$ 29,131
Net income applicable to common stockholders	\$ 27,410
Net income per common share data	
Basic earnings per share	\$ 8.89
Diluted earnings per share	8.88
Weighted-average basic shares	3,082.4
Weighted-average diluted shares	3,087.4

In order to go from net interest income to overall profitability, we need to account for fees. These are the fees here. They are listed under the bank's non-interest income, investment banking fees, profits and trading, fees charged on deposit accounts and mortgages. These could be overdraft fees or penalties for late payments, credit card fees, asset management fees. It's a long list, but we can take a closer look at some of the bigger items. They all come from the wholesale business.

Investment Banking Fees - JPM

Year ended December 31, (in millions)	2020
Advisory	\$ 2,368
Equity underwriting	2,758
Debt underwriting ^(a)	4,351
Total investment banking fees	\$ 9,477

(a) Represents long-term debt and loan syndications.

First, investment banking fees, M&A advisory, equity underwriting, bond and syndicated loan underwriting. JP Morgan earned around nine billion dollars in fees from this business line in 2020.

Proprietary Trading - JPM

Year ended December 31, (in millions)	2020
Trading revenue by instrument type	
Interest rate	\$ 2,575
Credit	2,753
Foreign exchange	4,253
Equity	6,171
Commodity	2,088
Total trading revenue	17,840
Private equity gains/ (losses)	181
Principal transactions	\$ 18,021

Prop trading, these are also called principal transactions. This is when JP Morgan's traders take speculative positions in financial markets to make profits on its own account. So this is not on behalf of its customers. This generated a whopping \$18 billions of profits in 2020.

Asset Management and Brokerage Fees - JPM

Year ended December 31, (in millions)	2020
Asset management fees	
Investment management fees	\$ 11,694
All other asset management fees	338
Total asset management fees	12,032
Total administration fees	2,249
Commissions and other fees	
Brokerage commissions	2,959
All other commissions and fees	937
Total commissions and fees	3,896
Total asset management, administration and commissions	\$ 18,177

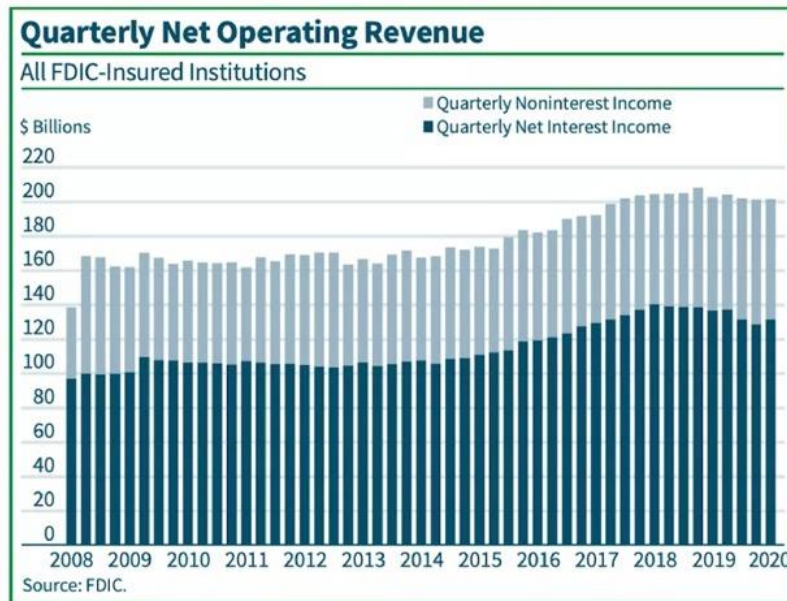
Then we have fees from asset management and brokerage services. This is when JP Morgan is helping its customers invest in financial markets. The bank earns asset management fees from managing portfolios. It also earns brokerage commission fees from helping with customer trades. This business line also generated \$18 billion in change. Adding up those fees gives us our non-interest income of \$64 billion.

Net Income - JPM

Year ended December 31, (in millions, except per share data)	2020
Revenue	
Investment banking fees	\$ 9,486
Principal transactions	18,021
Lending- and deposit-related fees ⁽¹⁾	6,511
Asset management, administration and commissions ⁽²⁾	18,177
Investment securities gains/(losses)	802
Mortgage fees and related income	3,091
Card income ⁽³⁾	4,435
Other income	4,457
Noninterest revenue	64,980
Interest income	64,523
Interest expense	9,960
Net interest income	54,563
Total net revenue	119,543
Provision for credit losses	17,480
Noninterest expense	
Compensation expense	34,988
Occupancy expense	4,449
Technology, communications and equipment expense	10,338
Professional and outside services	8,464
Marketing ⁽⁴⁾	2,476
Other expense	5,941
Total noninterest expense	66,656
Income before income tax expense	35,407
Income tax expense	6,276
Net income	\$ 29,131
Net income applicable to common stockholders	\$ 27,410
Net income per common share data	
Basic earnings per share	\$ 8.89
Diluted earnings per share	8.88
Weighted-average basic shares	3,082.4
Weighted-average diluted shares	3,087.4

For a bank as large and diversified as JP Morgan, this amount is actually bigger than the \$54 billions of profit generated by the lending business. Our last block of items gives us the non-interest expense. This includes salaries and bonuses, definitely the biggest item, rent, technology, as well as marketing and other smaller items. The bank also has to pay taxes. The net income of the bank takes the net interest income plus the non-interest income and subtracts these non-interest expense items. This is the bottom line of the bank. In 2020, it was a healthy \$30 billion. JP Morgan is a very large and modern bank that is diversified across multiple business lines. In 2020 at least, it had a large amount of fee income, so large in fact that its non-interest income was actually bigger than the net interest income generated by its lending business. Is JP Morgan a special case? Well, yes and no. Yes, it is the largest bank in the U.S. and more diversified than the average bank, and no because many other banks rely on fee income to support their business as well. The shift towards a fee business has been ongoing since the 1980s when the 3-6-3 model of banking came under intense pressure.

Net Income - All U.S. Banks



Here we can see the breakdown of net interest income and fee income for U.S. banks in recent times. For the average bank, about 1/3 of bank revenue comes from fees.

Summary

- Net interest income
- Fees

OK, let's wrap up. In this session, we covered the income statement. We saw how a modern bank generates profits from its lending business and also from fees. In the 21st century, banks are less reliant on interest income. In an environment with persistently low interest rates and new sources of competition in the loan market, focusing on fees makes a lot of sense. It is a form of diversification that allows banks to supplement lost interest income.